ENIR 11
ENERGY AND ENVIRONMENTAL ENGINEERING

Dr Aditya Kumar
Assistant Professor, Department of Energy & Environment
NIT Trichy
Email id: adityakumar@nitt.edu

Course Content

- Present Energy resources in India and its sustainability - Different type of conventional Power Plant--Energy Demand Scenario in India-Advantage and Disadvantage of conventional Power Plants – Conventional vs Non-conventional power generation
- Basics of Solar Energy- Solar Thermal Energy- Solar Photovoltaic- Advantages and Disadvantages-Environmental impacts and safety.
- Power and energy from wind turbines- India's wind energy potential- Types of wind turbines- Off shore Wind energy- Environmental benefits and impacts.
- Biomass resources-Biomass conversion Technologies- Feedstock preprocessing and treatment methods- Bioenergy program in India-Environmental benefits and impacts. Geothermal Energy resources –Ocean Thermal Energy Conversion – Tidal.
- Air pollution- Sources, effects, control, air quality standards, air pollution act, air pollution measurement. Water pollution-Sources and impacts, Soil pollution-Sources and impacts, disposal of solid waste.
- Greenhouse gases – effect, acid rain. Noise pollution. Pollution aspects of various power plants. Fossil fuels and impacts, Industrial and transport emissions- impacts.

Course Plan

S. No.	Weeks (Hours)	Topics/Units	Mode of Delivery
1	1 to 3 (6)	Unit I	Chalk & Talk, PPT
2	4 to 5 (3)	Unit II	Chalk & Talk, PPT
3	5 to 6 (3)	Unit III	Chalk & Talk, PPT
4	7 to 8 (4)	Unit IV	Chalk & Talk, PPT
5	9 to 10 (4)	Unit V	Chalk & Talk, PPT
6	11 to 12 (4)	Unit VI	Chalk & Talk, PPT

Text Books

- Boyle, G. 2004.' Renewable energy: Power for a sustainable future'. Oxford University press.
- B H Khan, 'Non Conventional Energy Resources'-The McGraw –Hill Second edition.
- G. D. Rai, 'Non conventional energy sources', Khanna Publishers, New Delhi, 2006.
- Gilbert M. Masters, 'Introduction to Environmental Engineering and Science', 2nd Edition, Prentice Hall, 2003.
- 'Unleashing the Potential of Renewable Energy in India' –World bank report.
- Godfrey Boyle, Bob Everett and Janet Ramage.2010.'Energy Systems and Sustainability. Power for a sustainable future'. Oxford Universitypress.

Course Assessment Methods

S. No.	Methods/Mode of Assessment	% Weightage
1	Mid Sem	20
2	Project idea submission	5
3	Objectives and Methodology submission	15
4	Results and report submission	30
5	Final Assessment	30

Course Objectives

- To introduce the energy consumption pattern in India and compare it with international scenario.
- To introduce the energy demand and potential of conventional energy resources of India.
- To educate on functioning of conventional power plants .
- To introduce various feedstocks of biomass along with its biochemical composition.
- To teach the different possible methods available for energy conversion of biomass.
- To introduce the function of ocean, tidal and geothermal energy power plants in terms of potential and energy cycles.
- To educate on the sources, composition, impacts and control measures of air pollution and water pollution
- To introduce the various source of industrial noise, its characteristics, measures, impacts, foot print and control methods.
- To discuss on green house gases, various impacts of fossil fuel use and pollutants from power plants
- To introduce the basics of solar energy and its applications in day-to-day life, and to teach the working principle of solar thermal and electrical devices.
- To introduce the advantages and disadvantages of use of solar energy, and to educate about the environmental impact of the solar devices and safety procedures to follow while handling it.
- To teach the working principle and basics of wind turbines, to introduce the wind energy potential and installed capacity of wind turbines in India.
- To introduce the different types of wind turbines used to harness the wind energy, and to introduce the offshore wind energy technology.
- To educate about the benefits of the wind energy and its impact on the environment.

Attendance Policy

- At least 75% attendance in each course is mandatory.
- A maximum of 10% shall be allowed under On Duty (OD)category.
- Students with less than 65% of attendance shall be prevented from writing the final assessment and shall be awarded 'V’ grade.

Unit I

Present Energy resources in India and its sustainability

- **As of 30 June 2025, India has 226.9 GW (49% of total) installed capacity of renewable energy. It is one of the world leaders in renewable energy investments and installations**

- India is simultaneously one of the world's fastest-growing energy consumers, a major coal-dependent power system, and one of the world's largest renewable-energy markets.

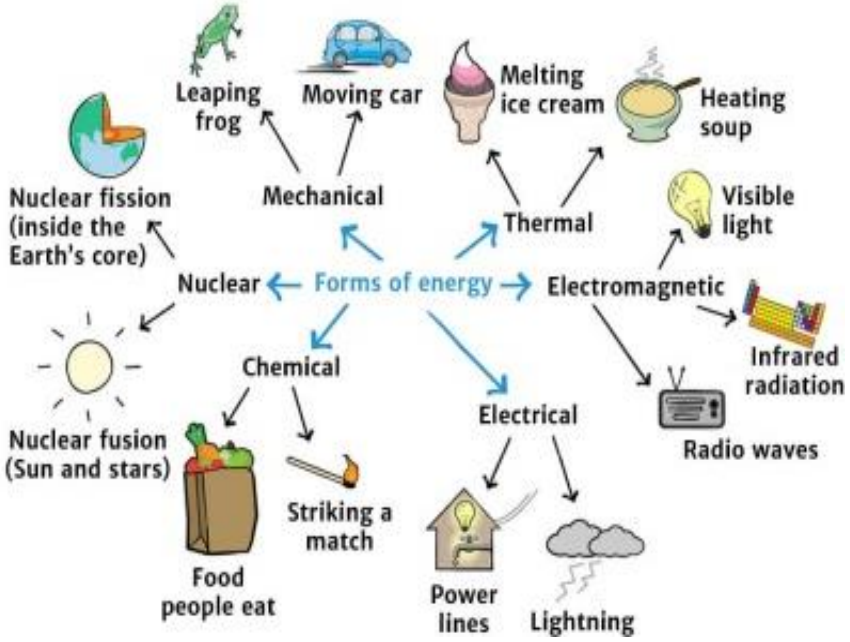
Present Energy resources in India and its sustainability

- **As of 30 June 2025, India has 226.9 GW (49% of total) installed capacity of renewable energy. It is one of the world leaders in renewable energy investments and installations.**
- **As of 31 July 2026, India has about 300.5 GW of non-fossil installed power capacity. MNRE currently reports that non-fossil sources account for roughly 54.45% of total installed capacity.**

“Does this mean that more than half of India's electricity now comes from renewable energy?”

Energy?

- Energy is the ***ability to cause changes***, exert forces or do work.
- In other words, energy is the ability to make things happen!
- An energy resource is something that can ***produce heat, power life, move objects, or produce electricity.***
- Matter that stores energy is called a fuel. Energy can be generated from fuel minerals like coal, Petroleum, natural gas, uranium and from electricity.



❑ **Sustainable energy** is the practice of using energy in a way that "*meets the needs of the present without compromising the ability of future generations to meet their own needs".*

Energy Units

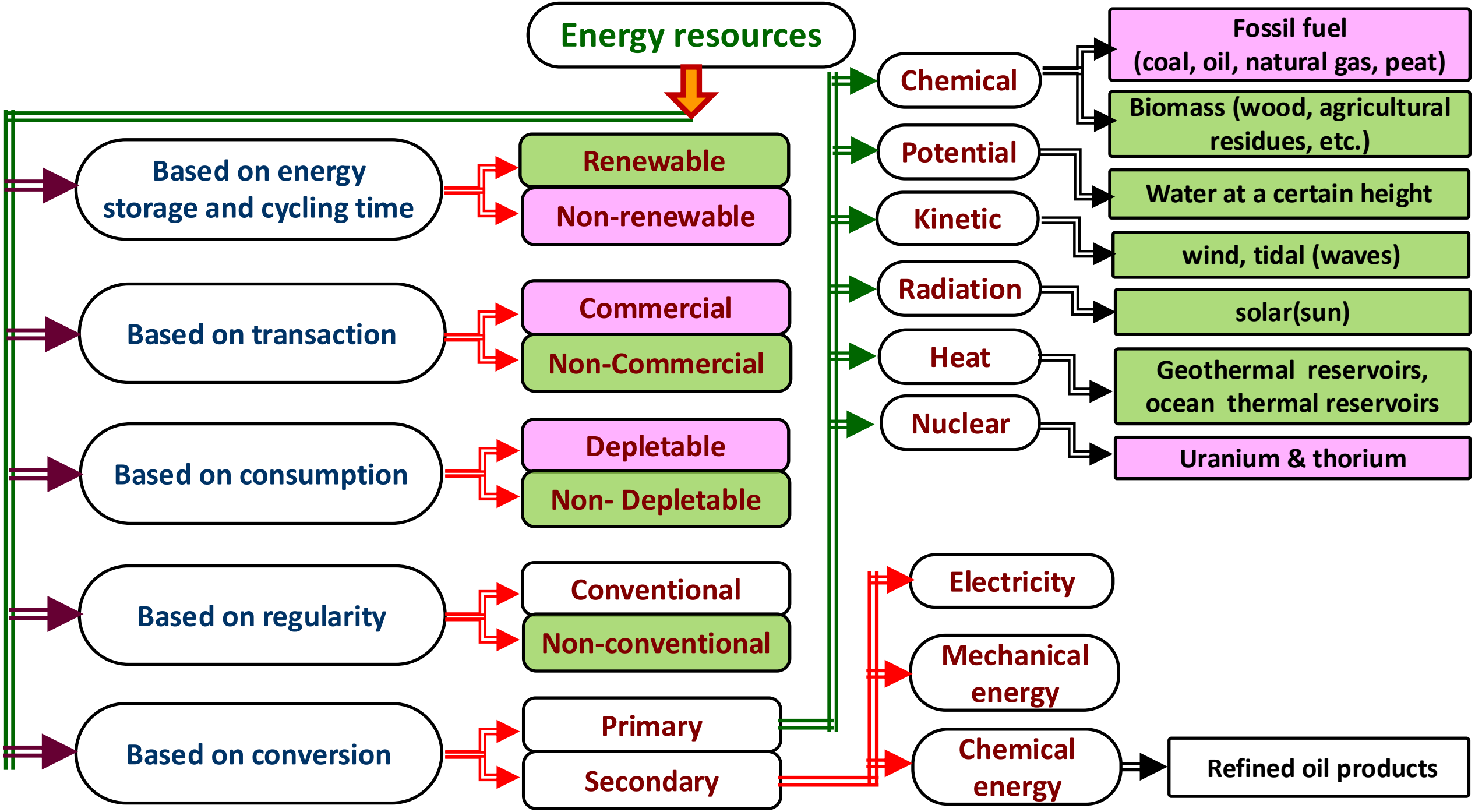
- SI unit of energy is the same as the unit of work – the joule (J), named in honor of James Prescott Joule.
- British Imperial units -
    - foot-pound force (1.3558 J)
    - British thermal unit (BTU) (1055 J)
    - horsepower-hour (2.6845 MJ)
    - gasoline gallon equivalent (about 120 MJ).
- Electricity- one kWh (3.6×10^6 J) (3600 kJ or 3.6 MJ).
- Natural gas- In US-sold in Therms /100 cubic feet (105.5 megajoules). In Australia-sold in Cubic Meters ($1m^3$ = 38 megajoules)
- Atom physics and chemistry- units electronvolts (eV)
- Food industry- a calorie of 4.184 J

Energy Units

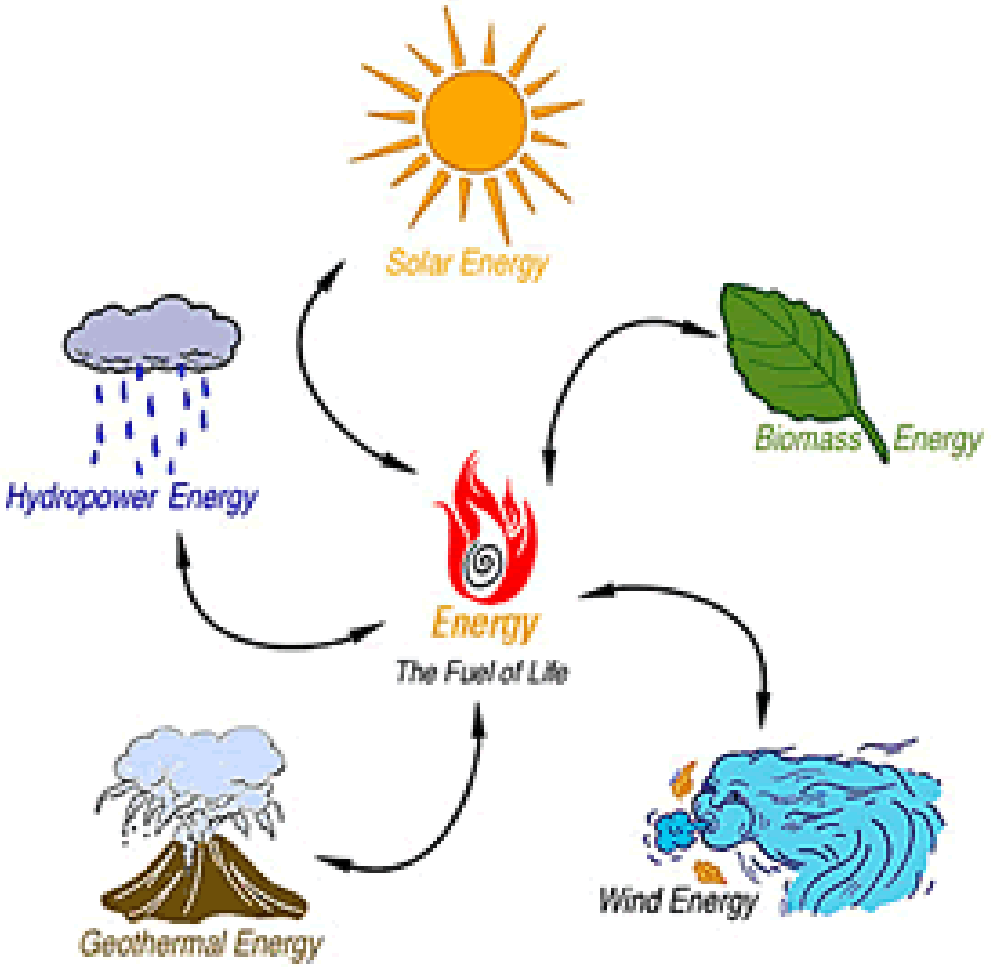
Energy conversion			
Unit	**Quantity**	**to**	**Note**
1 calorie =	4.1868000	Joule	
1 kiloWatt hour = kWh =	3600000	Joule	A power of 1 kW for a duration of 1 hour.
1 British Thermal Unit = btu	1055.06	Joule	It is a is a unit of energy used in North America.
1 ton oil equivalent = 1 toe	4.19E+010	Joule	It is the rounded-off amount of energy that would be produced by burning one metric ton of crude oil.
1 ton coal equivalent	2.93E+10	Joule	
1 ton oil equivalent = 1 toe	1 / 7.33	Barrel of oil	or 1 / 7.1 or 1 / 7.4 ...
1 cubic meter of natural gas	3.70E+07	Joule	or roughly 1000 btu/ft3
1000 Watts for one year	3.16E+010	Joule	for the 2000 Watt society
1000 Watts for one year	8.77E+006	kWh	for the 2000 Watt society
1 horsepower	7.46E+002	Watts	

Classification of Energy Resources

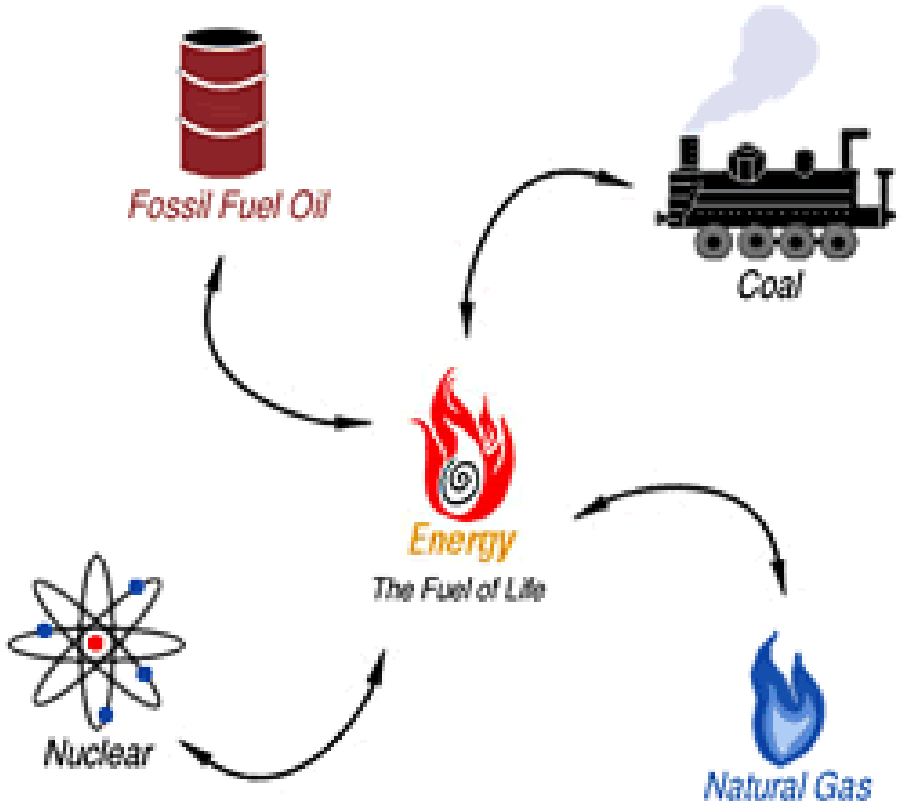
- Based on the **energy storage or cycling time**-Renewable or non-renewable resources (Renewable resources have a cycling time less than 100 years, while for non-renewable resources, it is greater than a million years)
- Based on the **nature of their transaction**- commercial -available to the users at some price (coal, petroleum, natural gas and electricity) and noncommercial sources- available free of cost to the users (fire wood, agricultural waste, cow dung)
- Based on **consumption**- exhaustible/ depletable- gets drained when used as an input of a production process and non - depletable / renewable resources
- Based on **regularity**-conventional -have been in use since a **long time** (coal, oil, hydro, nuclear, etc.) and non-conventional - recently developed and are still developing. (solar, wind, tidal, geothermal, biogas, etc.) sources.
- Based on **conversion**-primary or secondary types - (primary sources - found in nature that has not been subjected to any human engineered conversion process and secondary sources- made from other energy sources).


Energy resources
Based on energy storage and cycling time
Renewable
Non-renewable
Based on transaction
Commercial
Non-Commercial
Based on consumption
Depletable
Non- Depletable
Based on regularity
Conventional
Non-conventional
Based on conversion
Primary
Secondary
Chemical
Fossil fuel (coal, oil, natural gas, peat)
Biomass (wood, agricultural residues, etc.)
Potential
Water at a certain height
Kinetic
wind, tidal (waves)
Radiation
solar(sun)
Heat
Geothermal reservoirs, ocean thermal reservoirs
Nuclear
Uranium & thorium
Electricity
Mechanical energy
Chemical energy
Refined oil products

Renewable Energy

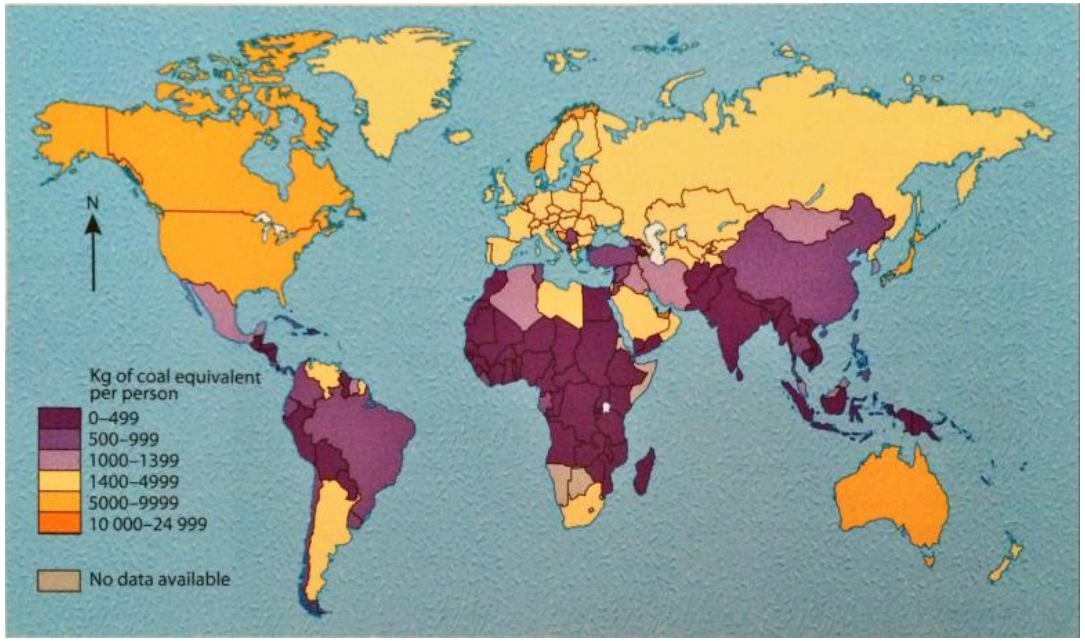


Non-Renewable Energy

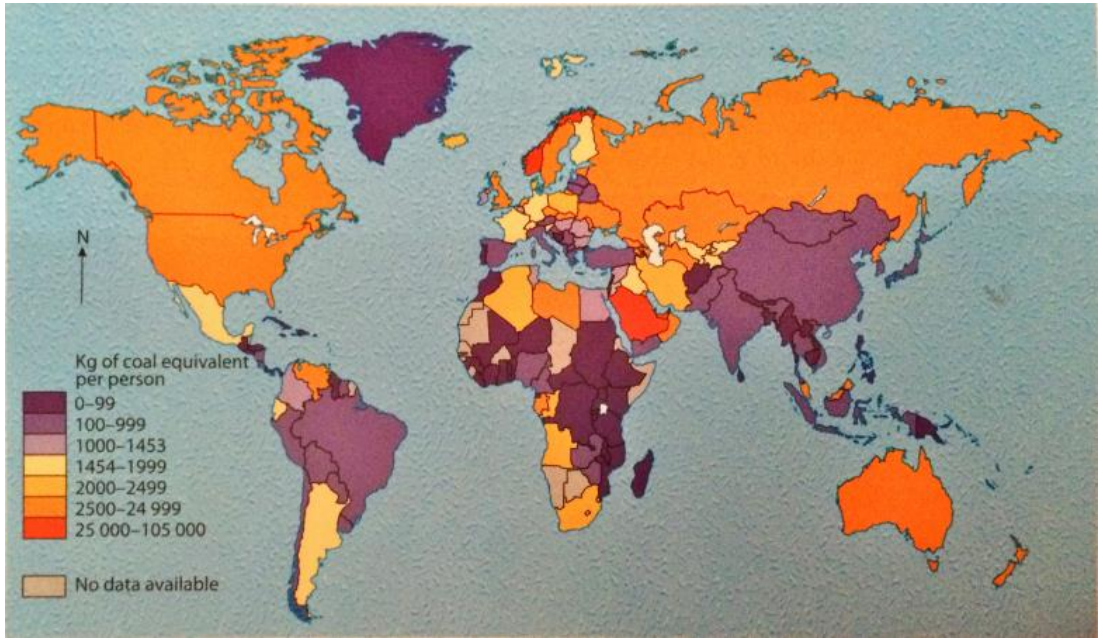


Global report on energy demand

Energy consumers
(Kg of coal equivalent to per person)



Energy producers
(Kg of coal equivalent to per person)



- **Europe & North America use 70 % of the world energy, although this is only 20% of the world's population-experienced large-scale economic development, there original energy sources were fossil fuels – first coal and the oil and gas**
- **Demand for energy over much of South America, Africa & South-east Asia is very low-less economically developed**

- **World's major consumers of energy are also the major producers i.e. the countries with less energy demand and consumption have less energy production but still the resources to have a higher energy production.**
  - Global **electricity** demand is projected to grow between 62 and 185 percent by 2050 compared with 2021 levels. The share of **fossil fuels** in the electricity mix declines from 59 percent in 2021 to 2–55 percent by 2050

Energy gap

- Energy gap - the difference between a country's rising demand for energy and its ability to produce that energy from its own resources.
- The gap is being widened by the deliberate phasing out of fossil fuels.
- The loss of energy is greater than the amount of energy being generated for other renewable sources.
- The mismatch between the distribution of energy consumption and Production.

India Energy Statistics

Year	Requirement	Availability	Surplus(+)/Deficts(-)		Peak Demand	Peak Met	Surplus(+) / Deficts(-)	
2020-21	12,75,534	12,70,663	-4,871	-0.4	1,90,198	1,89,395	-802	-0.4
2021-22	13,79,812	13,74,024	-5,787	-0.4	2,03,014	2,00,539	-2,475	-1.2
2022-23	7,91,053	7,85,722	-5,331	-0.7	2,15,888	2,07,231	-8,657	-4.0
2023-24	1560485	1536858	-23627	-1.5	2,43,000	2,35,000	-8,000	-3.3
2024-25	1624000	1598000	-26000	-1.6	2,56,500	2,48,000	-8,500	-3.3
2025-26*	1700000	1675000	-25000	-1.5	2,70,000	2,61,000	-9,000	-3.3

If India's peak demand is 300 GW and India has 500 GW of installed capacity, can India still have a power shortage?

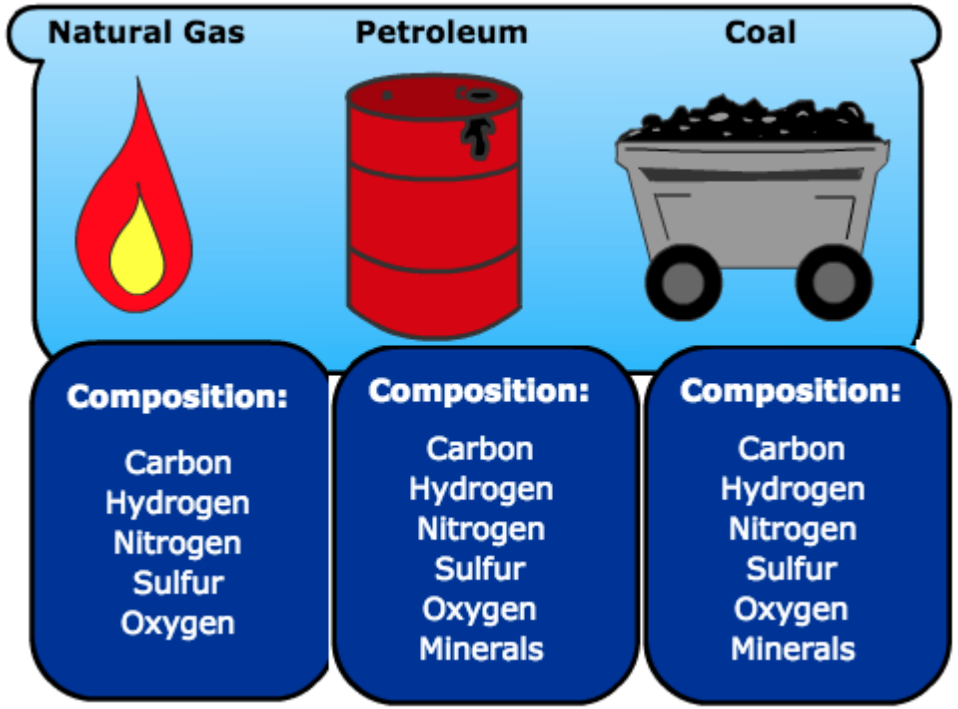
Non-Renewable Energy Sources

- Fossil Fuels: Fossil fuels such as coal , oil and natural gas all of which are mixture of compounds containing carbon and hydrogen. These are formed by the decomposition of the remains of plants and animals buried under the earth. Millions of years ago.
  - Crude Oil : petroleum products include *gasoline, distillates such as diesel fuel and heating oil, jet fuel, petrochemical feedstocks, waxes, lubricating oils, and asphalt.*
  - Natural Gas: Natural gas consists mainly of *methane*; it can be used as a fuel or to make materials and chemicals.
  - Coal: It's a sedimentary rock with a high amount of *carbon and hydrocarbons*.
- Uranium (Nuclear Energy): Most widely used by nuclear plants for nuclear fission. Uranium is considered to be a nonrenewable energy source, even though it is a common metal found in rocks worldwide. Uranium, referred to as U-235.

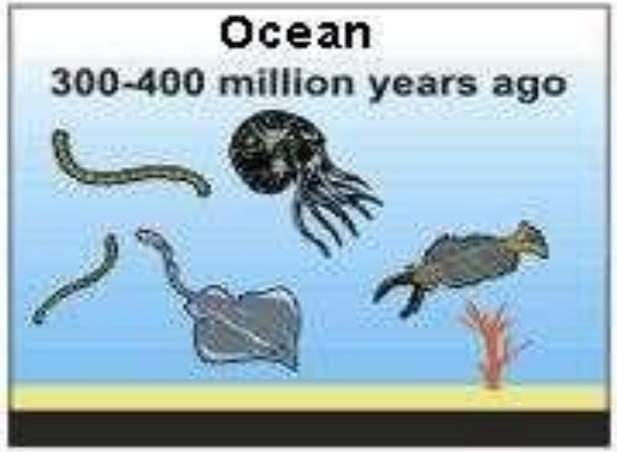
FOSSIL FUEL

- Fossil fuels are hydrocarbons comprised primarily of the following elements: carbon and hydrogen and some sulfur, nitrogen, oxygen, and mineral matter. Mineral matter turns into ash when burnt.

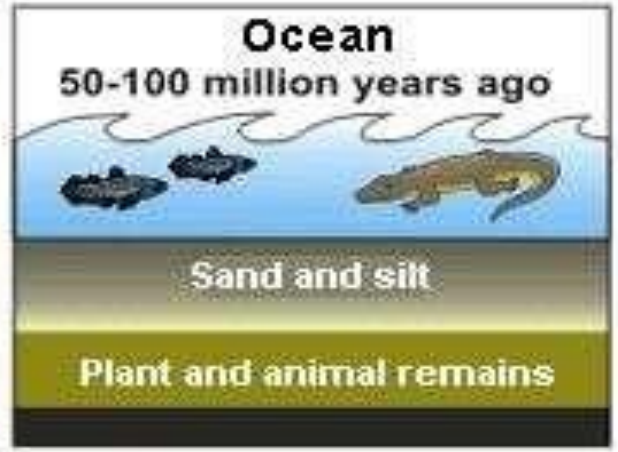
- The ***composition and the amounts of these elements change for different fossil fuels*** (coal, petroleum, and natural gas), but the elements are the same. For example, there is more hydrogen in liquid fuels than in coal per unit mass.



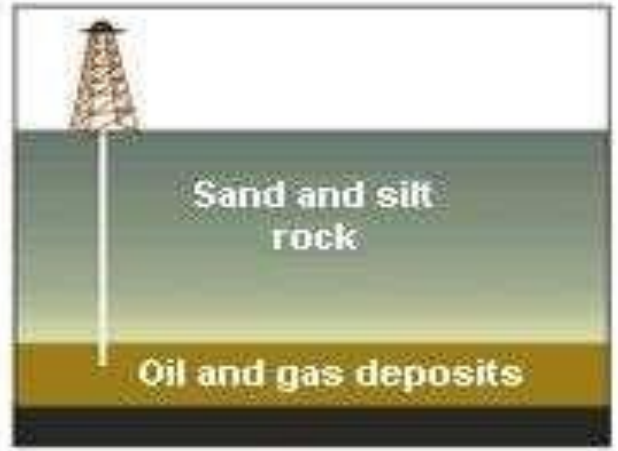
Petroleum and natural gas formation



Tiny sea plants and animals died and were buried on the ocean floor. Over time, they were covered by layers of silt and sand.

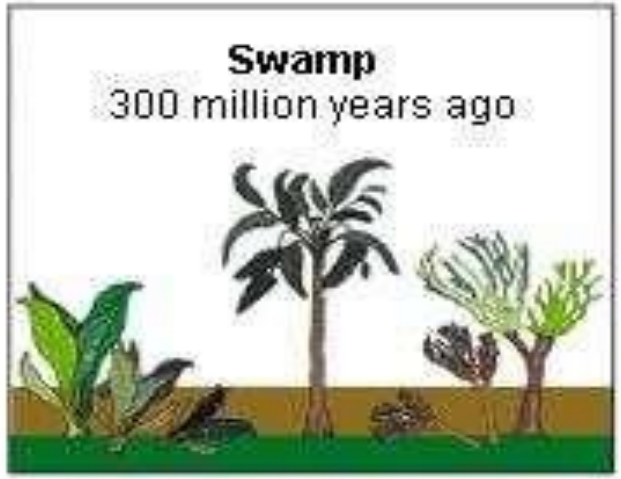


Over millions of years, the remains were buried deeper and deeper. The enormous heat and pressure turned them into oil and gas.

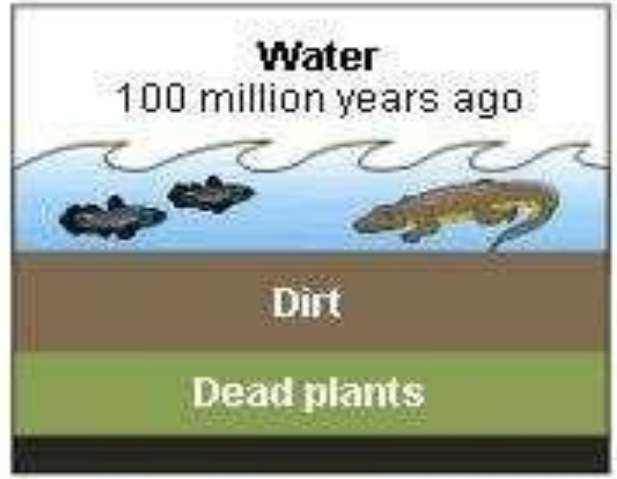


Today, we drill down through layers of sand, silt, and rock to reach the rock formations that contain oil and gas deposits.

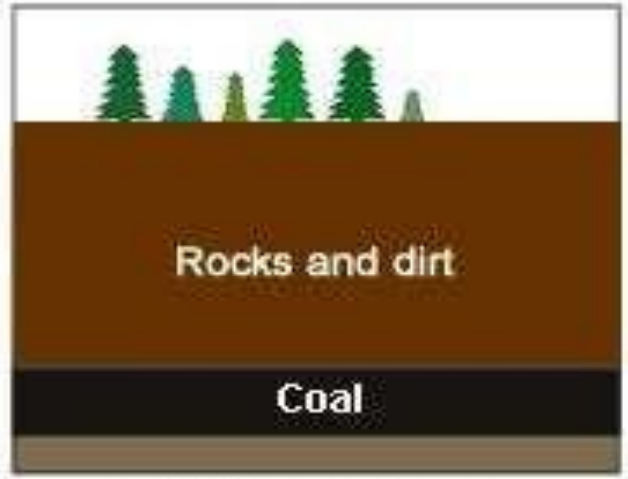
How coal was formed



Before the dinosaurs, many giant plants died in swamps.

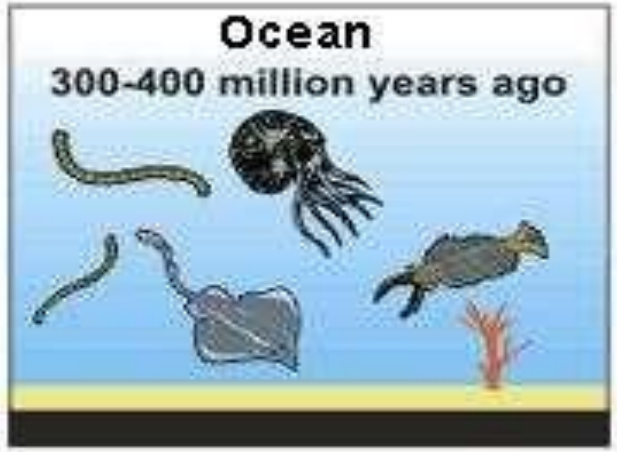


Over millions of years, the plants were buried under water and dirt.

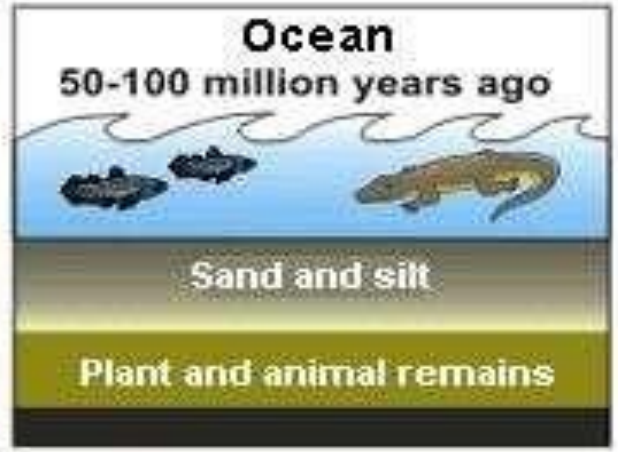


Heat and pressure turned the dead plants into coal.

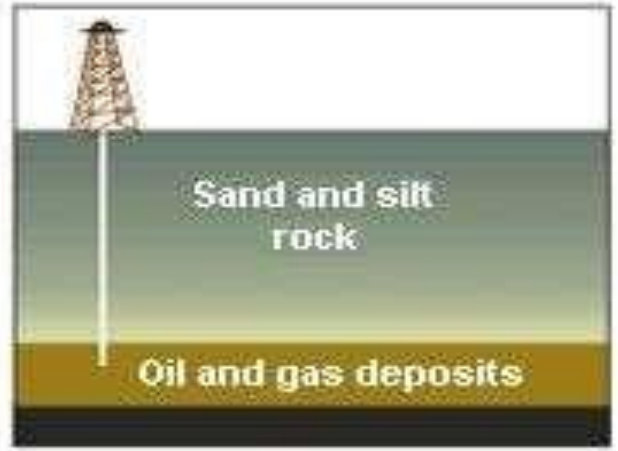
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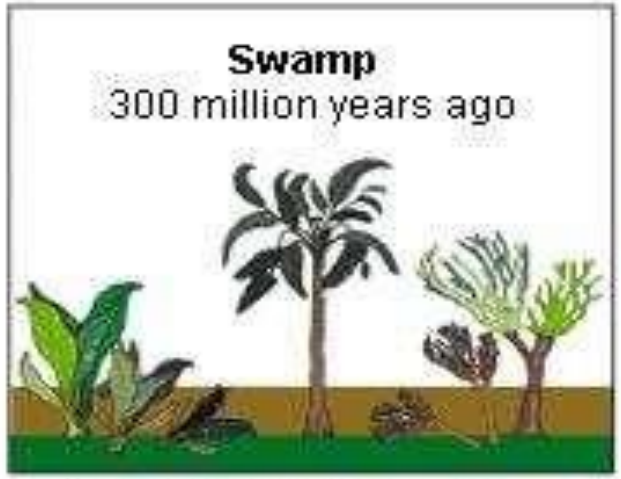


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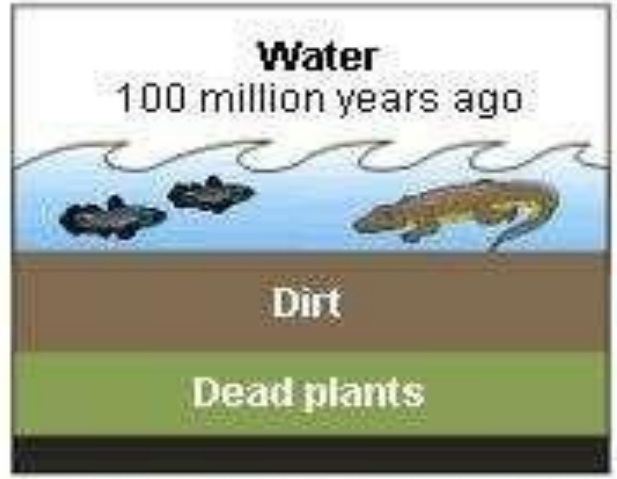


Today, we drill down through layers of sand, silt, and rock to reach the rock formations that contain oil and gas deposits.

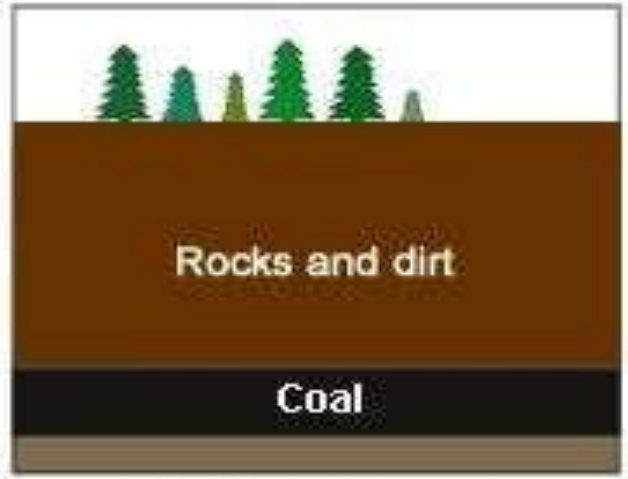
How coal was formed



Before the dinosaurs, many giant plants died in swamps.



Over millions of years, the plants were buried under water and dirt.



Heat and pressure turned the dead plants into coal.

Coal

- Coal is the main source of energy in India as it fulfils almost *over 50 % of the total commercial energy consumed* in the country. India is the ***2nd largest producer of coal,*** *after China.*
- This fossil fuel is found in a form of sedimentary rocks and is often known as ***'Black Gold'***.
- Coal is formed due the compression of plant material over millions of years. It is originated from organic matter wood. When large tracts of forests are buried under sediments, wood is burnt and decomposed due to heat from below and pressure from above. The phenomenon makes coal but takes centuries to complete.
- It is found in a ***variety of forms depending on the degrees of compression and the depth and time of burial***. Decaying plants in swamps produce peat. Which has a low carbon and high moisture contents and low heating capacity.



Classification of Coal

Classification on the basis of carbon content and time period.

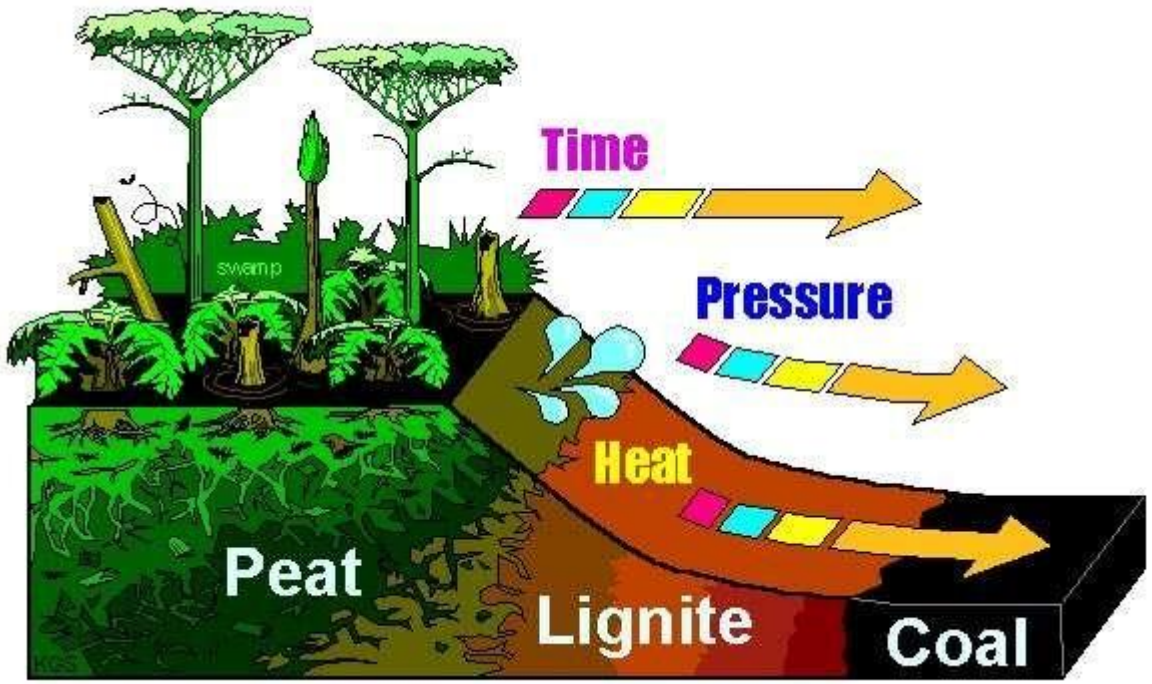
Types of coal on the basis of carbon content

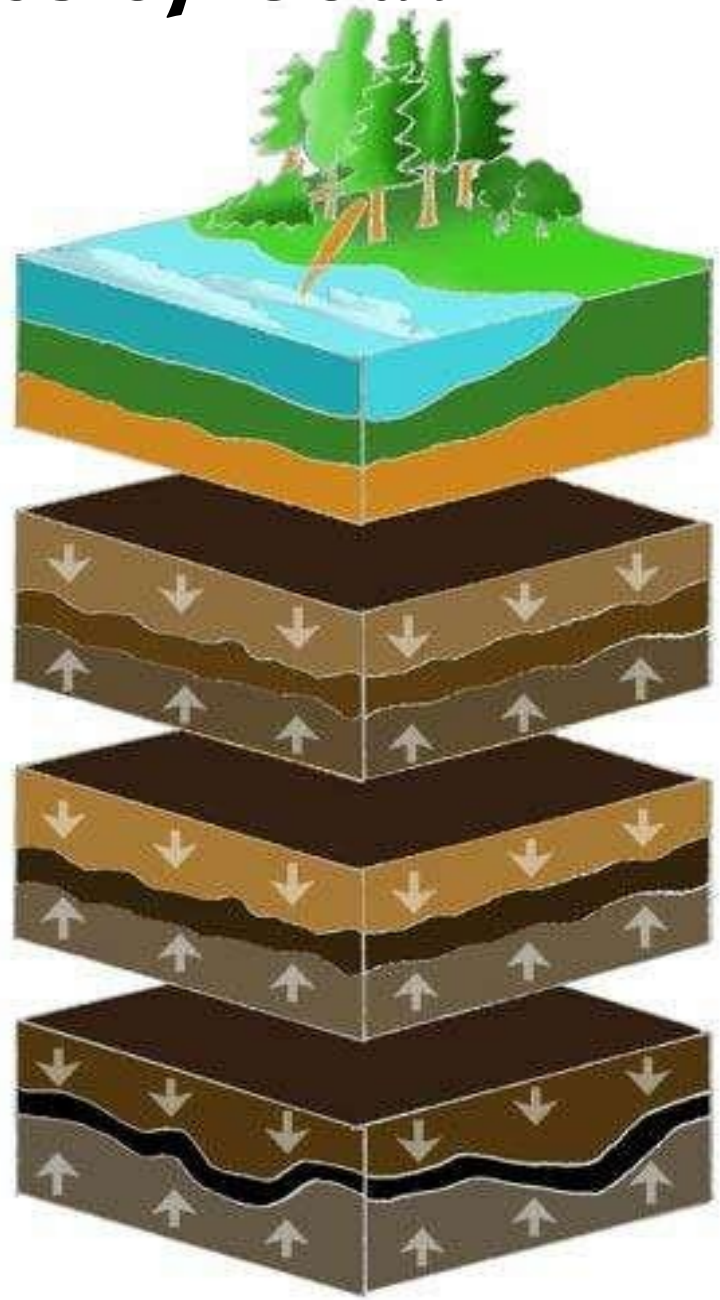
- *Anthracite (hard coal)*-- best quality of coal which carries 80-95 % C content. It ignites slowly with a blue flame. It has the *highest calorific value*. (Jammu and Kashmir).
- *Bituminous (Soft coal)*-- 60-80 % C content and a *low level of moisture content*. It is widely used and has *high calorific value*. (Jharkhand, West Bengal, Odisha, Chhattisgarh and Madhya Pradesh).
- *Lignite (brown coal)*-- 40-55 % C content. It is an intermediate stage which happens during the alteration of woody matter into coal. It has *high moisture content* so it gives smoke when burnt. (Rajasthan, Lakhimpur (Assam), and Tamil Nadu).
- *Peat* -- less than 40 % C content. It is in the first stage of transformation from wood to coal. It has *low calorific value* and burns like wood.

Types of coal on the basis of a time period

- *Gondwana coal:* Around 98 % of India's total coal reserves are from Gondwana times. This coal was formed about 250 million years ago.
- *Tertiary coal* is of younger age. It was formed from 15 to 60 million years ago.

Four Stages of Coal





HUGE FORESTS GREW AROUND 300 MILLION YEARS AGO COVERING MOST OF THE EARTH

THE VEGETATION DIES AND FORMS PEAT

THE PEAT IS COMPRESED BETWEEN SEDIMENT LAYERS TO FORM LIGNITE

FURTHER COMPRESSION FORMS BITUMINOUS AND SUBITTUMINOUS COAL

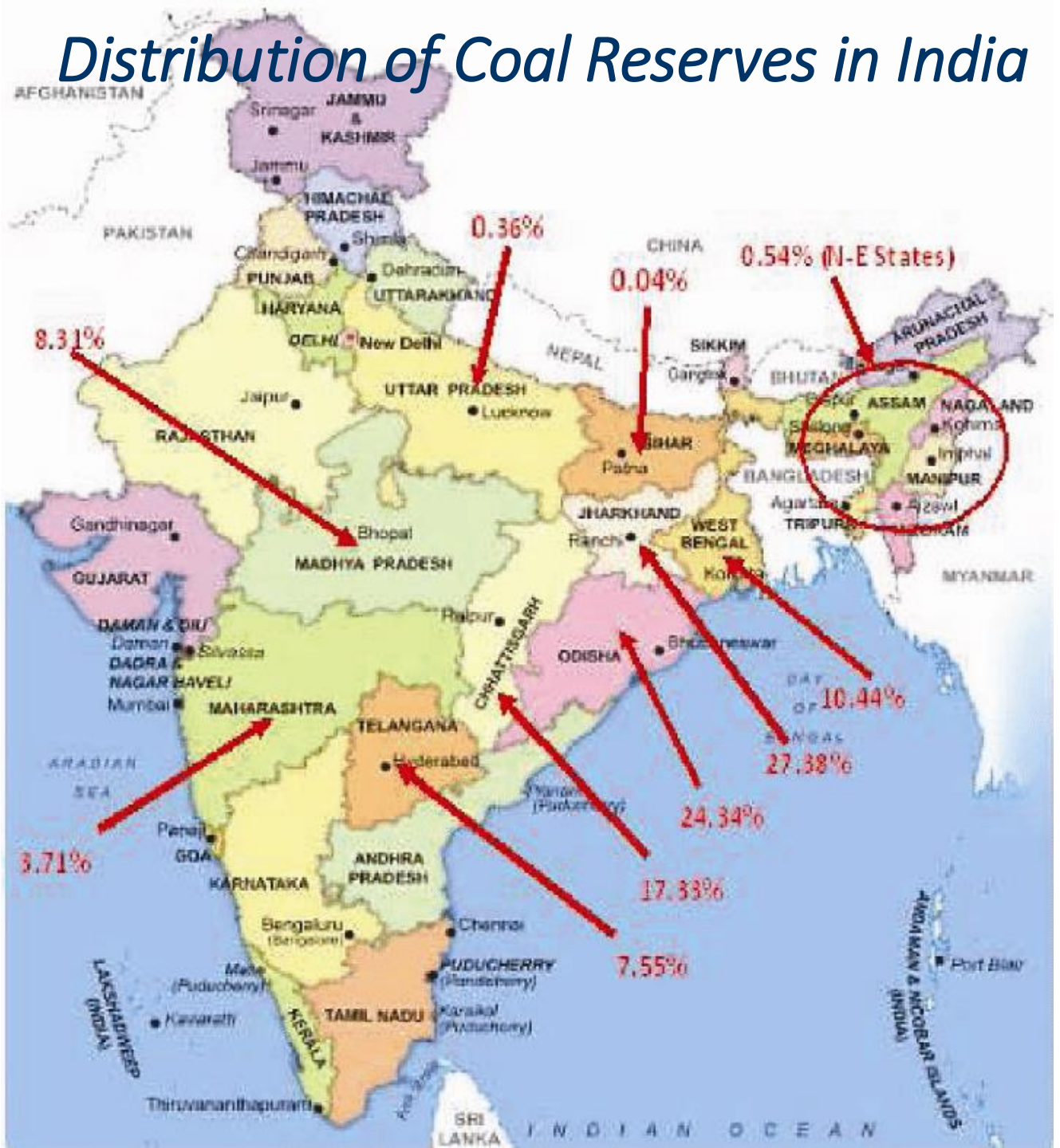
EVENTUALLY ANTHRACITE FORMS

Coal reserves in India: State-wise

State	Type of coal	Coal field with Coal mining centers	Total coal reserves
Odisha		Talchar (Dhenkanal and Sambalpur), Ib Valley (Jharsuguda)	25.0%
Jharkhand	Bituminous (Gondwana period)	Jharia(Dhanbad), Bokaro (Hazaribagh), Auranga, Giridh, Karanpur, Ramgarh and Hutar.	23.2%
Chhattisgarh		Korba(valley of river Hasdo), Chirmiri, Jhimli, and Johilla	21.3%
West Bengal	bituminous	Raniganj (Bardhman & Birbhum), Darjeeling, Bankura, Jalpaiguri, and Puruliya	9.0%
Madhya Pradesh		Singrauli, Muhpani, Satpura, Pench Kanhan and Sohagpur	8.5%
Maharashtra	more moisture and has less carbon content	Kamptee (Nagpur) and Wardha valley (Nagpur and Yavatmal)	3.5%
Andhra Pradesh		Singareni (Godavari valley)	1.1%
Rajasthan	Lignite	Palana and Khari mines (Bikaner)	
Gujarat	poor quality 35% of carbon	Bharauch and Kachchh	
Tamil Nadu	lignite coal	Neyveli	
Jammu and Kashmir	inferior quality	Shaliganga, Handwara, Baramulla, Riasi, and Udhampur	

- As of **1 April 2025**, India's total identified geological coal resource is about **400.72 billion tonnes (BT)**, according to the National Coal Inventory reported by the Ministry of Coal/GSI.

Distribution of Coal Reserves in India



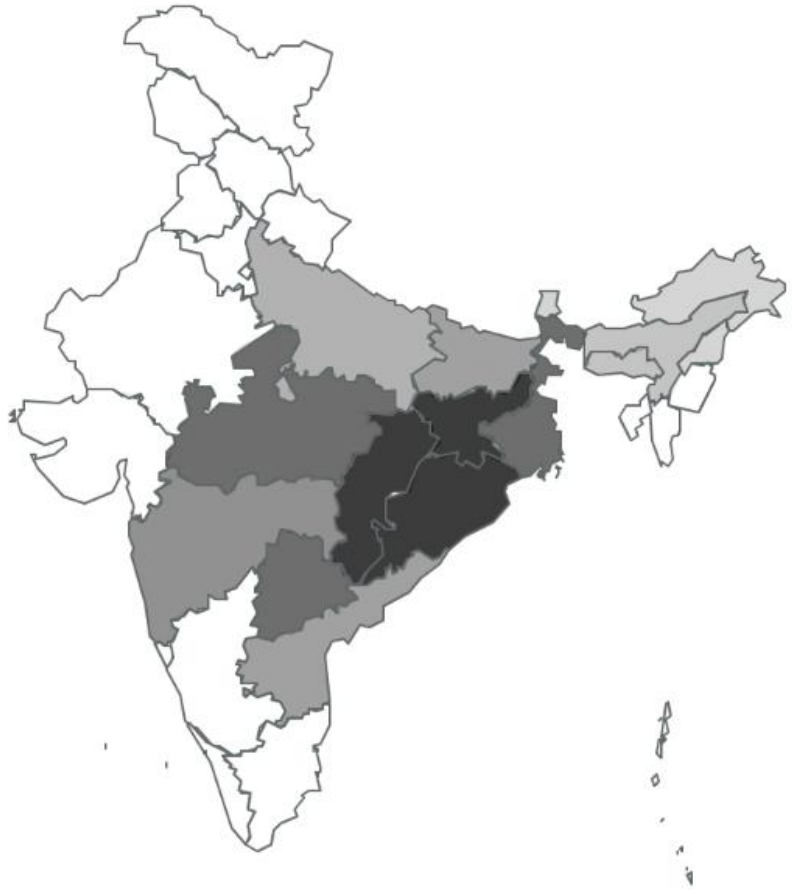
❑ Advantages

- The coal reserves are huge.
- The cost of production and transport is less.
- The low quality coal can be used for production of tar.

❑ Dis-Advantages

- It creates air and water pollution.
- It is heavy and bulky, so storing needs large areas.
- The ash produced on burning of coal creates health problems.

Coal Reserves in India



Quantity in Million Tonnes

389,421

Quantity in Million Tonnes

- 0-500
- 500-1000
- 1000-2000
- 2000-8000
- 8000-16000
- 16000-20000
- 20000-50000
- 50000-60000
- 60000-80000
- 80000-100000

Source: MoC

Lignite Reserves in India



Quantity in Million Tonnes

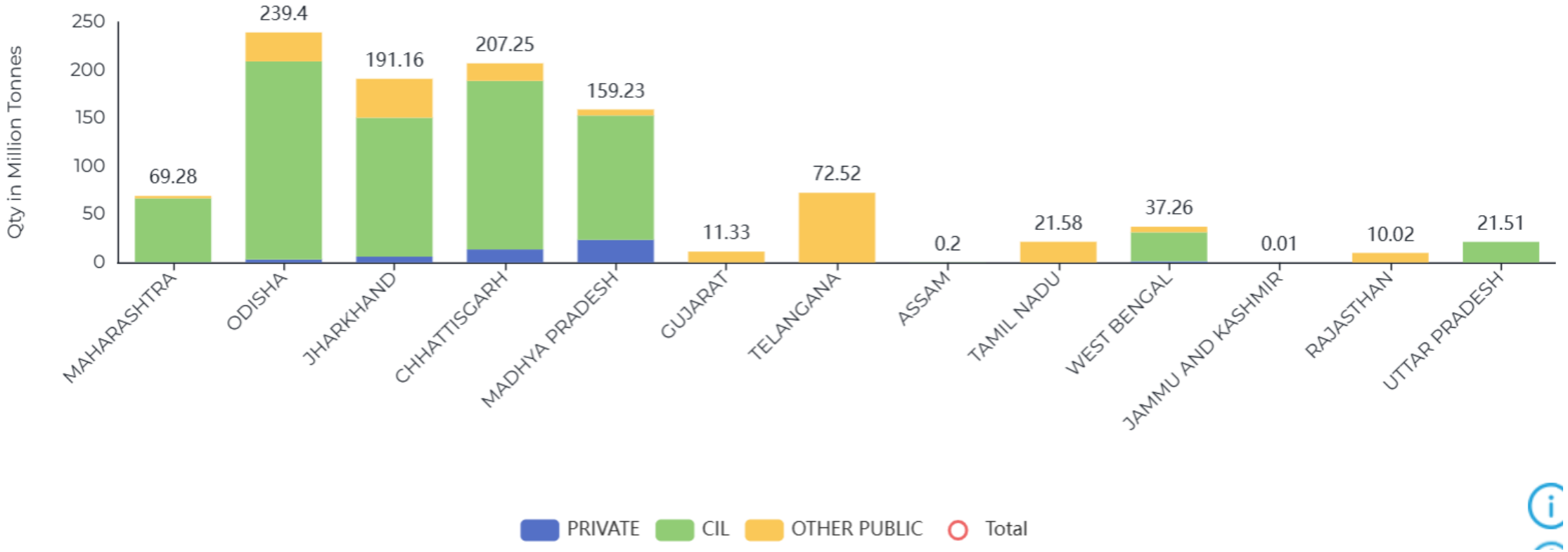
47,296

Quantity in Million Tonnes

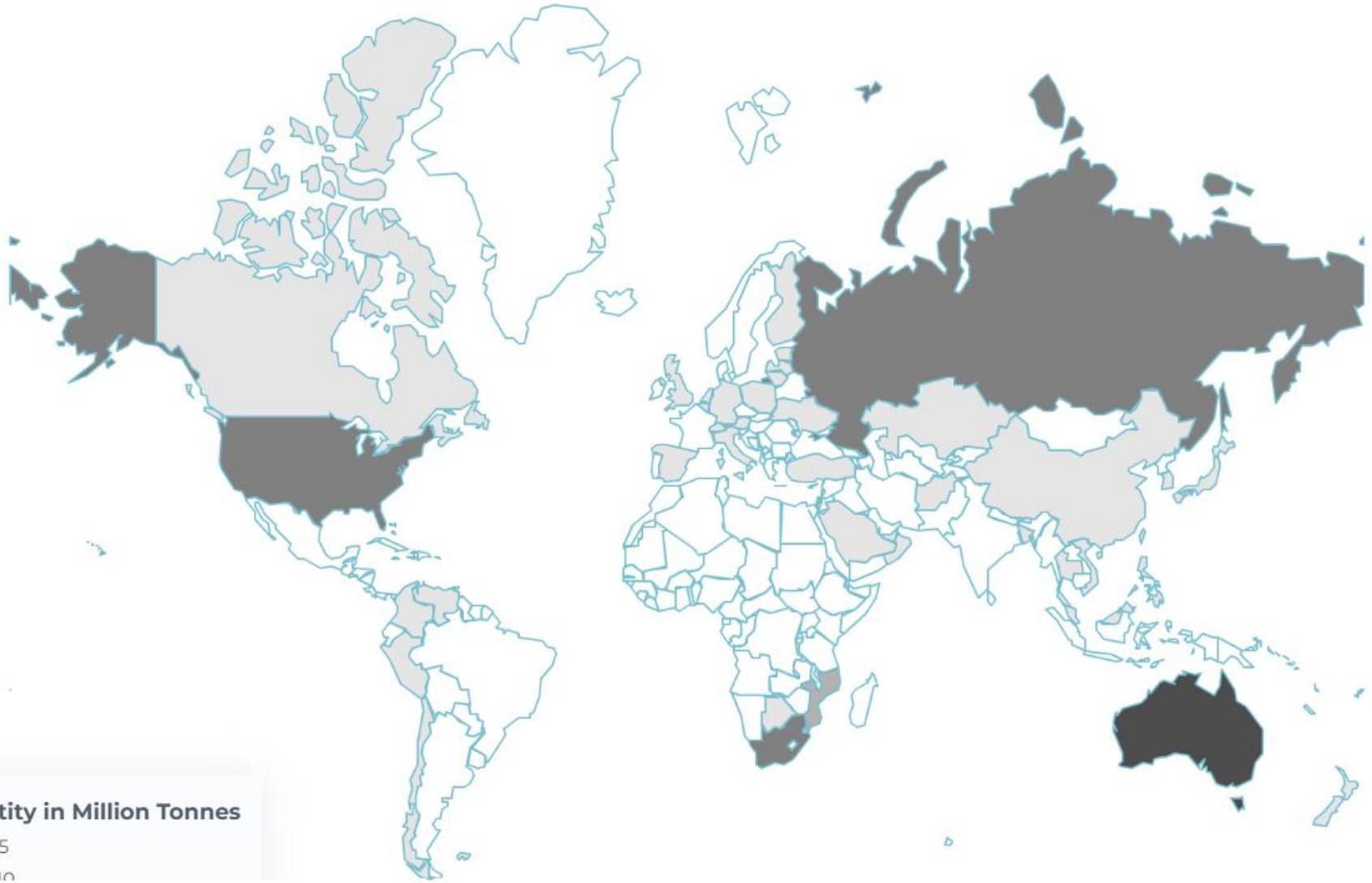
- 0-500
- 500-1000
- 1000-2000
- 2000-8000
- 8000-16000
- 16000-20000
- 20000-50000
- 50000-60000
- 60000-80000
- 80000-100000

Source: MoC

Production of Coal in India



Import of Coal to India



Amount (in Crores)

317800.37

Quantity (in Million Tonnes)

264.90

India currently produces roughly **1.04 billion tonnes of coal per year**.
(FY2025-26)

Why does India import coal despite huge production?

If India has ~401 billion tonnes of geological coal resources and mines ~1 billion tonnes every year, can we simply say India has coal for 400 years?

CRUDE OIL



- **CRUDE OIL**: Crude oil means a mixture of hydrocarbons that exists in liquid phase in natural underground reservoirs and remains liquid at atmospheric pressure after passing through surface separating facilities.

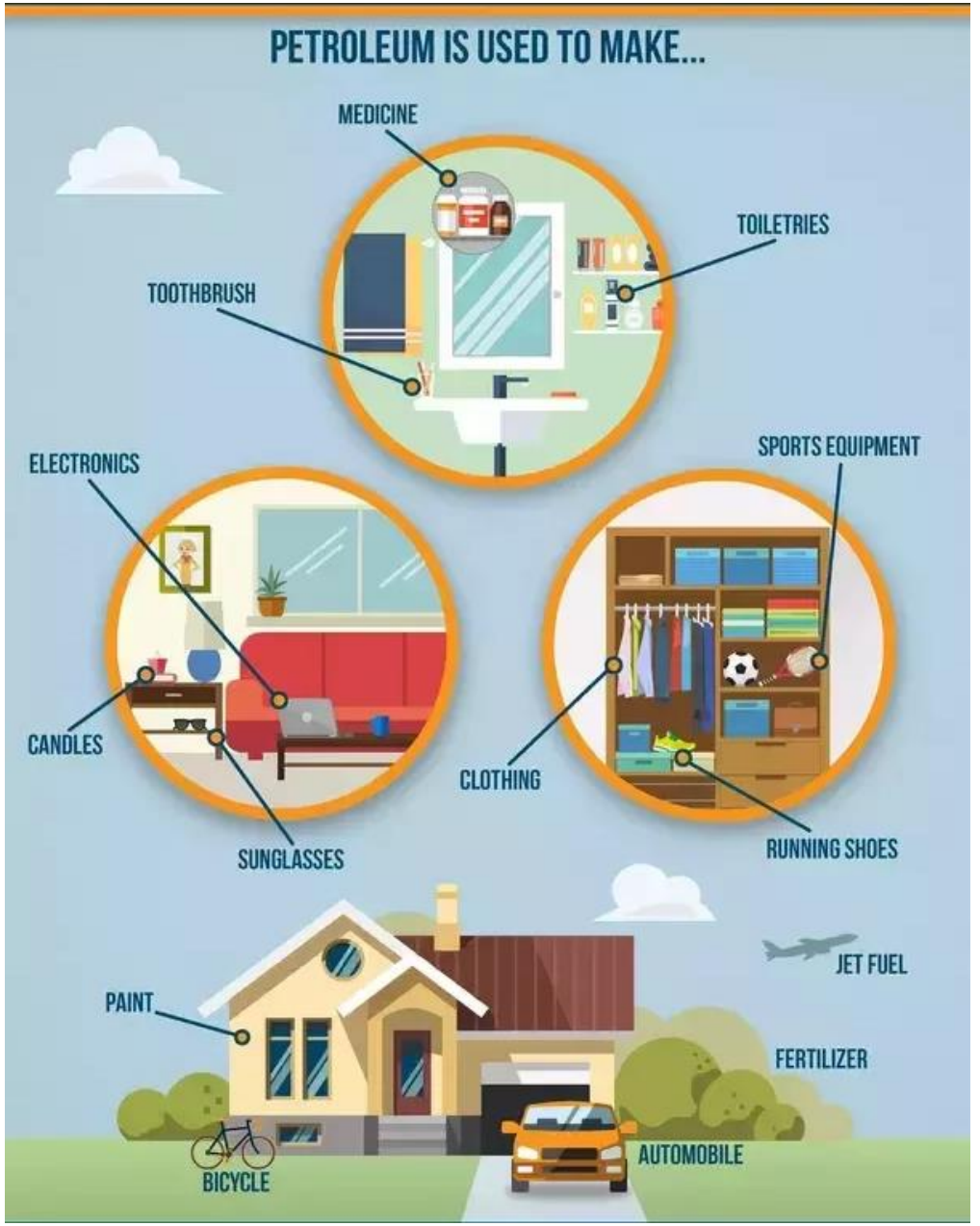
- **PETROCHEMICALS**: chemicals derived from petroleum, essential component of over 3000 products, products include: tar, wax, and plastic.

Petroleum

- 'Petroleum' -Latin words Petra (meaning rock) and Oleum (meaning oil). Thus petroleum is oil obtained from rocks; particularly sedimentary rocks of the earth. Therefore, it is also called mineral oil.
- Petroleum is an inflammable liquid that is composed of hydrocarbons which constitute 90-95 % of petroleum and the remaining is chiefly organic compounds containing oxygen, nitrogen, sulphur and traces of organo-metallic compounds.
- When ancient animals were buried in prehistoric oceans and lakes heat and pressure build up. chemical changes eventually convert the remains into petroleum and natural gas.
- Petroleum is the next major energy source in India after coal. Petroleum refineries act as a "nodal industry" for synthetic textile, fertilizer and numerous chemical industries.
- Crude petroleum consists of a mixture of hydrocarbons-solid, liquid and gaseous. These include compounds belonging to the paraffin series and also some unsaturated hydrocarbons and small proportion belonging to the benzene group.

Utilization of Petroleum

- Petroleum and petroleum products are mainly used as motive power. It is a compact and convenient liquid fuel which has revolutionized transportation on land, in the air and on water. It can be easily transported from the producing areas to the consuming areas with the help of tankers and more conveniently, efficiently and economically by pipelines.
- It emits very little smoke and leaves no ash, (as is the case in coal utilisation) and can be used upto the last drop. It provides the most important lubricating agents and is used as an important raw material for various petro-chemical products.

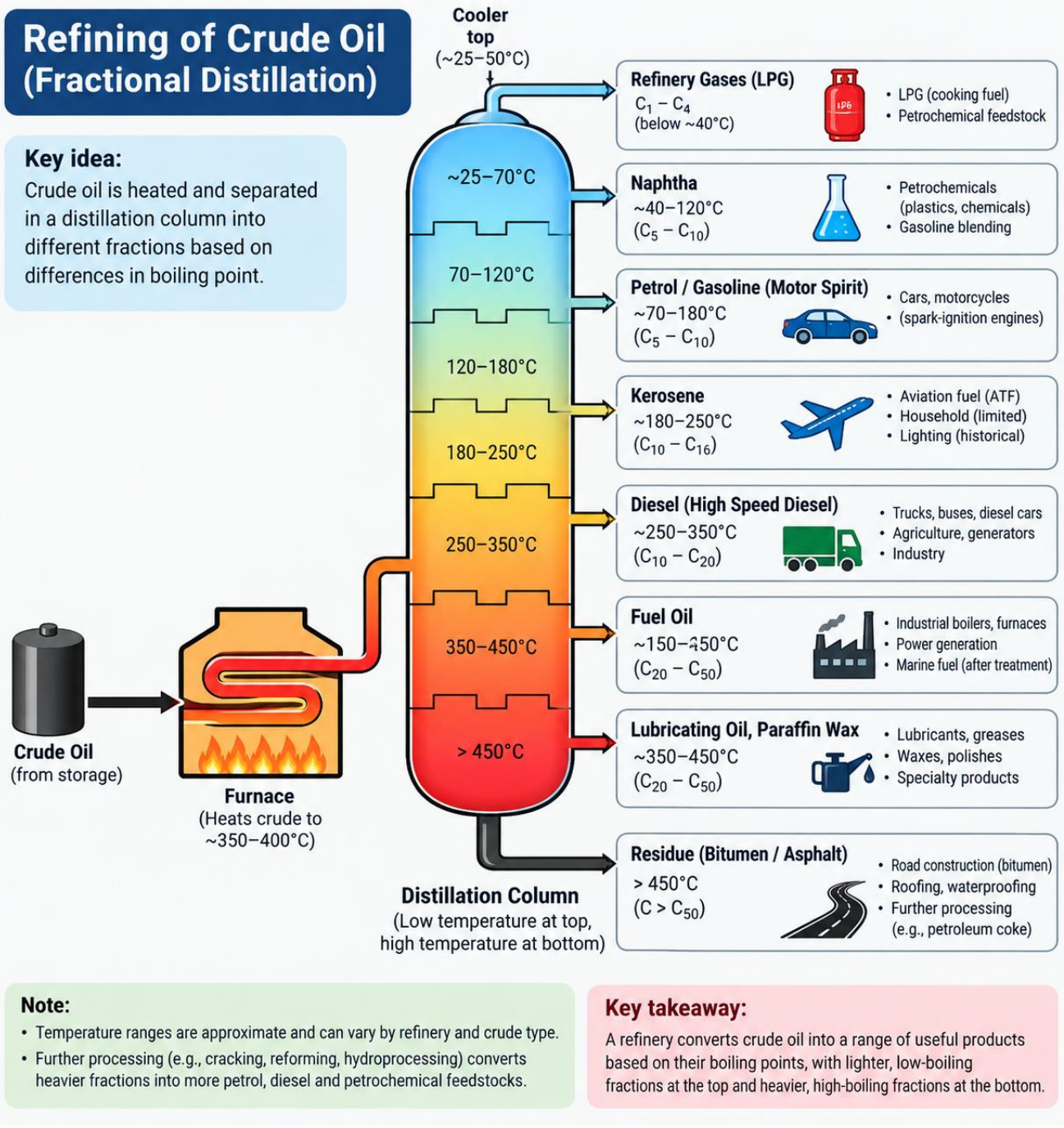


Origin and Occurrence of Petroleum

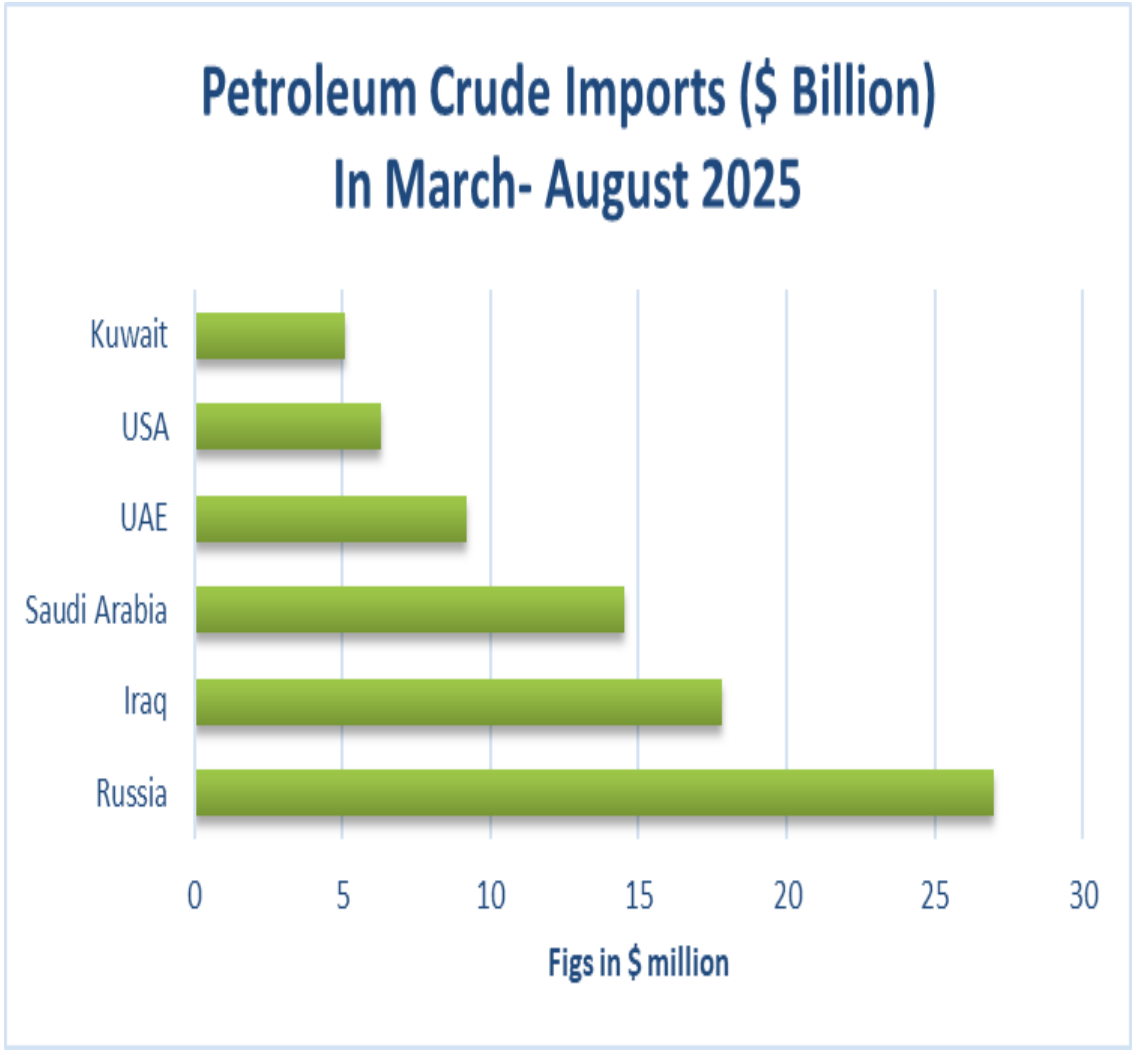
- Petroleum has an organic origin and is found in **sedimentary basins, shallow depressions and in the seas** (past and present). Most of the oil reserves in India are associated with anticlines and fault traps in the sedimentary rock formations of tertiary times, about 3 million years ago.
- Conditions for oil formation were favorable especially in the **lower and middle Tertiary period**. Dense forests and sea organisms flourished in the gulfs, estuaries, deltas and the land surrounding them during this period.
- The decomposition of organic matter in the sedimentary rocks has led to the formation of oil.
- An oil reservoir must have **three necessary conditions**:
  - porosity so as to accommodate sufficiently large amounts of oil;
  - permeability to discharge oil and/or gas when well has been drilled;
  - the porous sand beds sandstone, combination of fissured limestone containing oil should be capped by impermeable beds so that oil does not dissipate by percolation in the surrounding rocks.
- About 50 % of India's petroleum production is from Mumbai, 16 % from Gujarat and 14 % from Assam.

Crude oil → Refinery → LPG + Petrol + Naphtha + ATF/Kerosene + Diesel + Fuel oil + Lubricants + Bitumen + Petrochemical feedstocks

Product	Major utilization
Petrol / Motor Spirit (MS)	Cars, motorcycles and other spark-ignition vehicles
Diesel / HSD	Trucks, buses, diesel cars, agricultural machinery, generators, industry
LPG	Domestic cooking, commercial and industrial heating
ATF / Jet fuel	Aviation
Naphtha	Petrochemicals, chemicals and industrial feedstock
Kerosene	Domestic/industrial uses; aviation-related kerosene fractions
Fuel oil / LSHS	Industrial boilers, furnaces and process heating
Lubricants & grease	Engines, turbines, gears and industrial machinery
Bitumen	Road construction and waterproofing
Petroleum coke	Cement and selected industrial applications



India Petroleum liquid imports



	Coal	Crude oil
India's domestic availability	Very large	Limited
Import dependence	Relatively lower	~89%
Why imports?	Mainly quality/specific coal requirements + supply	Insufficient domestic crude production
Main consumption	Electricity + industry/steel	Transport + petrochemicals
Major strategic concern	Environment + reliability	Energy security + import bill + geopolitics + environment

Petroleum

- **Advantages**
  - The **transport** of liquid fuel is easy.
  - The liquid fuel is **lighter than coal** so it can be used in air transport.
  - The liquid fuel has **lots of bi-products**.
  - The liquid fuel does not require **very huge area of storing**.
- **Disadvantages**
  - The fuel is evaporable and so it is to be **stored in closed tankers**
  - The **production cost is higher** than that of coal.
  - It **creates air pollution** through the pollutants like NO, SO_2, CO, CO_2.
  - The **reserves of fuel oil are lesser** than that of coal

Strategic Petroleum Reserves (SPR) – India

Emergency crude-oil buffer to strengthen energy security during major supply disruptions

PHASE I

5.33 MMT

operational crude storage

≈ 9.5 days of crude requirement*

Visakhapatnam

1.33 MMT

Andhra Pradesh

Underground rock caverns

Mangaluru

1.50 MMT

Karnataka

Underground rock caverns

Padur

2.50 MMT

Karnataka

Underground rock caverns

PHASE II – Expansion

Additional 6.5 MMT planned on PPP / commercial-cum-strategic model

- Padur-II, Karnataka: 2.5 MMT — target completion Aug 2030
- Chandikhol, Odisha: 4.0 MMT — land/project development in progress

WHY SPR MATTERS

- India imports ~88–89% of its crude oil
- Buffers war, sanctions, shipping & supply shocks
- Government gets emergency access to strategic crude

Takeaway: SPR is an insurance policy for oil supply — it does not reduce import dependence, but buys time during disruption.

*9.5-day estimate is based on 2019–20 consumption. Sources: PIB / Ministry of Petroleum & Natural Gas; ISPRL Annual Report 2024–25.

What is India doing to reduce the risk?

Think of India's strategy as multiple parallel pathways:

- **Diversify crude suppliers** — India now sources from more than 40 countries.
- **Increase domestic exploration and production** — especially offshore/deepwater opportunities.
- **Strategic Petroleum Reserves** — emergency buffer against temporary supply disruptions.
- **EV adoption** — substitute electricity for petrol/diesel in road transport.
- **Ethanol blending** — substitute domestically produced biofuel for part of petrol demand.
- **Public transport and efficiency** — reduce petroleum consumption per passenger-km or tonne-km.
- **Alternative fuels** — biofuels, CNG/biogas and potentially hydrogen in suitable sectors.

India imports almost 90% of its crude. Then why does India export petrol and diesel?

Natural Gas

- Natural gas is an important **clean energy resource found in association with or without petroleum**. It is used as a source of energy as well as an industrial raw material in the petrochemical industry. Natural gas is considered an environment friendly fuel because of **low carbon dioxide emissions** and is, therefore, the fuel for the present century.
- Large reserves of natural gas have been discovered in the **Krishna- Godavari basin**. Along the west coast the reserves of the **Mumbai** High and allied fields are supplemented by finds in the **Gulf of Cambay**. **Andaman and Nicobar islands** are also important areas having large reserves of natural gas



- Consists primarily of **methane;** *Propane, butane, pentane, and hexane* are also present.
- Liquefied petroleum gas (LPG)-- Mixture of **butane and propane.**
- Commonly occurs in association with crude oil.
- Natural gas is often found dissolved in oil or as a gas cap above the oil.
- Oil + Gas -- Associated Gas – Wet Gas (pressure of natural gas forces oil up to the surface)
- Only Gas -- Non-Associated Gas – Dry Gas,
- Hydrogen Sulphide in gas-- Sour Gas
- Coalbed Methane == Sweet Gas (lack of hydrogen sulfide).
- On the market, natural gas is usually bought and sold not by volume but by calorific value.
- In practice, purchases of natural gas are usually denoted as MMBTUs (millions of British thermal unit (BTU or Btu)) = ~1,000 cubic feet of natural gas.

Natural Gas Formation

- Natural gas was formed millions of years ago when plants and tiny sea animals were buried by sand and rock.
- Layers of mud, sand, rock, plant, and animal matter continued to build up until the pressure and heat turned them into oil and natural gas.

Uses of Natural Gas

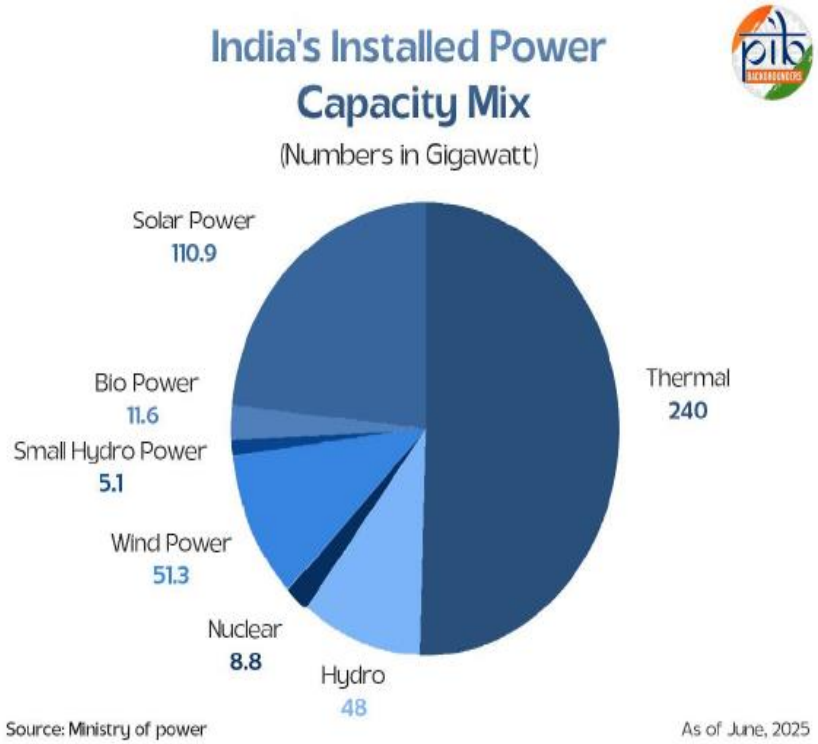
- Electric power generation.
- Industrial, domestic, and commercial usage.
- Many buses and commercial automotive fleets now operate on CNG.
- It is an ingredient in dyes and inks .
- Used in rubber compounding operations.
- Ammonia is manufactured using hydrogen derived from methane. Ammonia is used to produce chemicals such as hydrogen cyanide, nitric acid, urea, and a range of fertilizers.

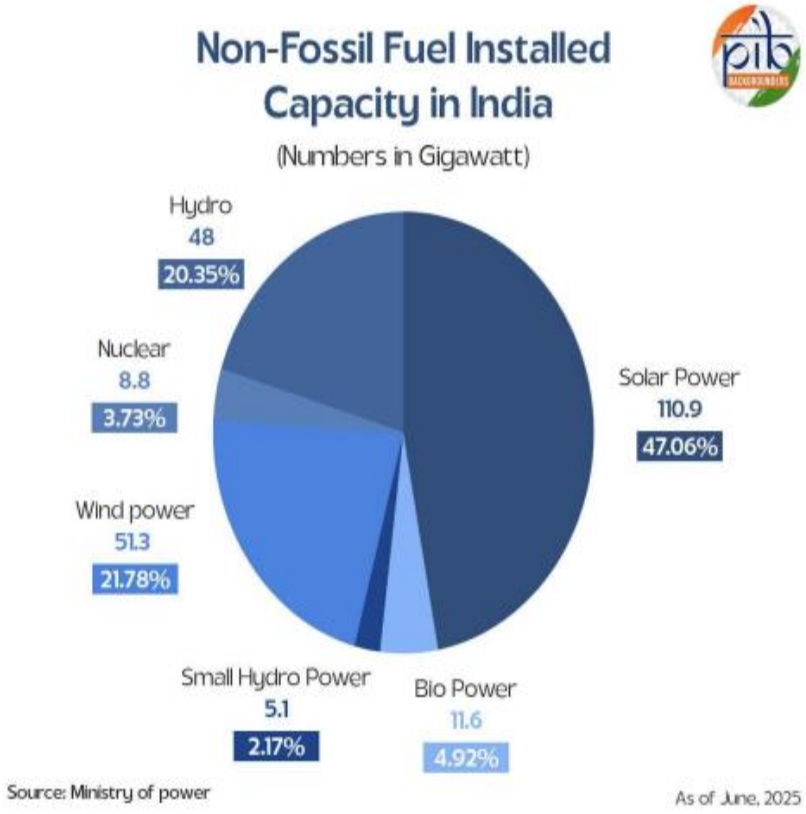
Importance of Natural Gas to India

- Power stations using gas accounted for nearly 10 per cent of India's electricity.
- Existing plants are operating below capacity on expensive imported liquefied natural gas (LNG).

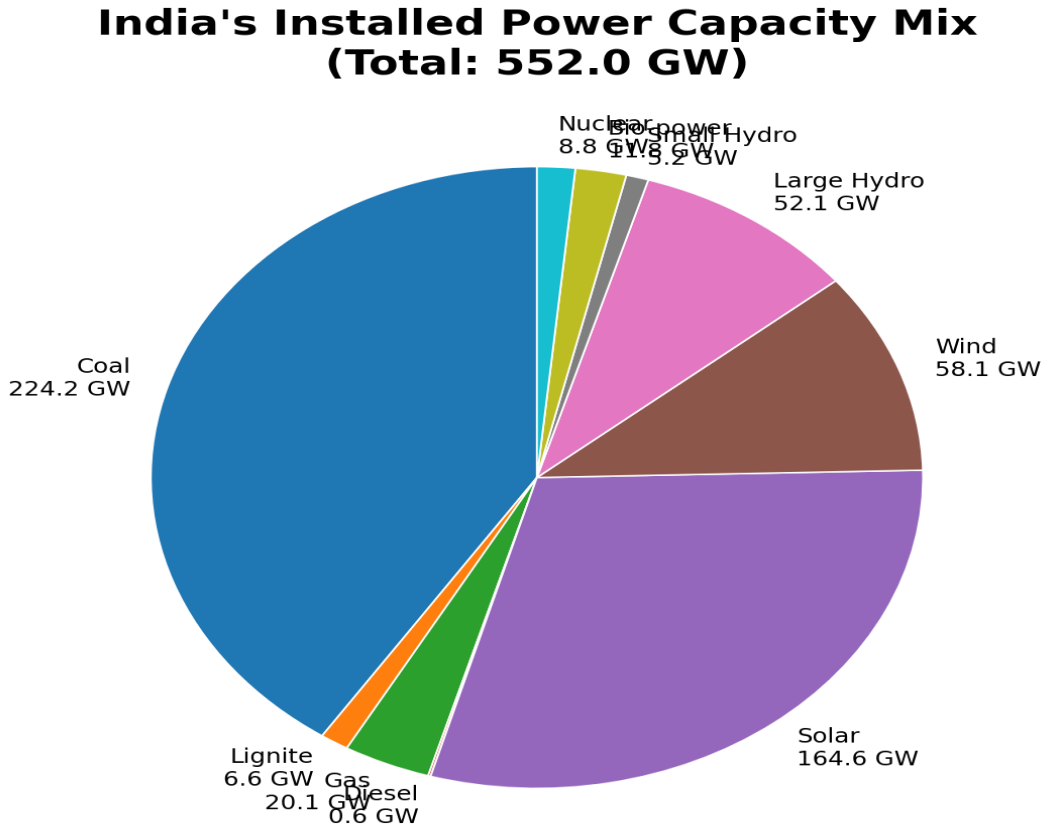
	Coal	Crude Oil	Natural Gas
India's domestic availability	High	Low	Moderate
Import dependence	Relatively low	~89%	~50%
Major use	Electricity/industry	Transport + petrochemicals	Fertilizer + CGD + industry + power
CO_2 intensity	Highest	Intermediate	Lower
Main import form	Coal	Crude oil	LNG
Storage/transport	Relatively easy	Relatively easy	More difficult
Energy-transition role	Declining over long term	Transport dependence reducing	Bridge fuel

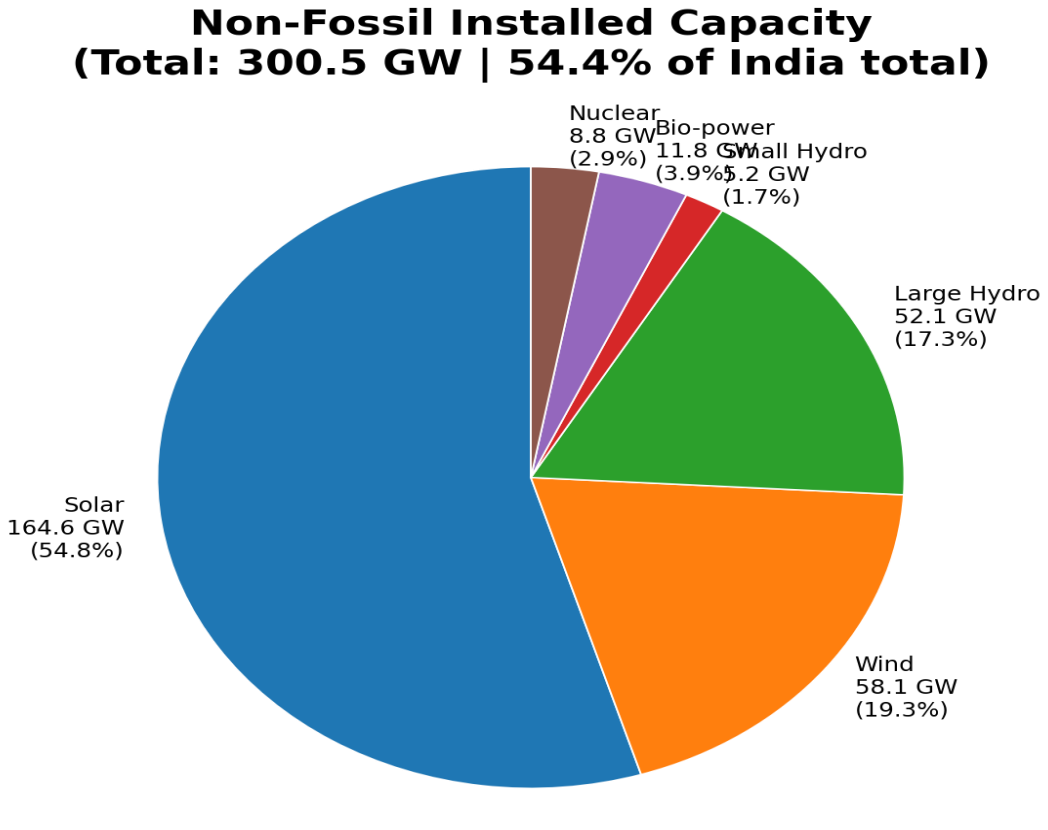
Total Installed Capacity (As on 30.06.2025) - Source : Ministry of Power





Total Installed Capacity (As on 31.07.2026) - Source: CEA / MNRE





Key shift: Non-fossil sources now exceed fossil sources in installed capacity — 300.5 GW (54.4%) vs 251.5 GW (45.6%).

Sources: Central Electricity Authority, Installed Capacity Report – July 2026; MNRE Physical Achievements as on 31 July 2026.

Installed GENERATION CAPACITY(FUELWISE) AS ON 30.06.2025		
CATAGORY	**INSTALLED GENERATION CAPACITY(MW)**	**% of SHARE IN Total**
Fossil Fuel		
Coal	2,19,000	46.01%
Lignite	711	0.15%
Gas	20,000	4.2%
Diesel	589	0.12%
Total Fossil Fuel	**2,40,300**	**50.48%**
Non-Fossil Fuel		
RES (Incl. Hydro)	**226,900**	47.67%
Hydro	**48,000**	**10.08 %**
Wind	51,300	10.78 %
Solar	1,10,900	23.30 %
BM Power/Cogen	11,600	2.44 %
Small Hydro Power	5,100	1.07 %
Nuclear	8,800	1.85%
Total Non-Fossil Fuel	**2,26,900**	47.67%
Total Installed Capacity (Fossil Fuel & Non-Fossil Fuel)	**4,76,000**	**100%**

Installed Generation Capacity (Fuel-wise) as on 31.07.2026

CATEGORY	INSTALLED CAPACITY (MW)	SHARE OF TOTAL
Fossil Fuel		
Coal	224,157.50	40.61%
Lignite	6,620.00	1.20%
Gas	20,122.42	3.65%
Diesel	589.20	0.11%
Total Fossil Fuel	**251,489.12**	**45.56%**
Non-Fossil Fuel		
Solar	164,594.75	29.82%
Wind	58,136.89	10.53%
Large Hydro	52,064.67	9.43%
Small Hydro	5,181.76	0.94%
Bio-power + Waste-to-Energy	11,747.57	2.13%
Nuclear	8,780.00	1.59%
Total Non-Fossil Fuel	**300,505.64**	**54.44%**
Total Installed Capacity	**551,994.76**	**100.00%**

Sources: CEA Installed Capacity Report, July 2026; MNRE Physical Achievements as on 31 July 2026.
Note: Bio-power + Waste-to-Energy combines biomass/cogeneration and waste-to-energy categories. Figures may differ slightly due to rounding.

Different type of conventional power plant

POWER PLANT

- A power plant is an industrial facility used to generate electric power with the help of one or more generators which converts different energy sources into electric power.

- Electricity is a secondary energy source, which means that electricity is obtained from the conversion of other primary sources of energy, such as coal, natural gas, nuclear, solar, or wind energy. The energy sources used to make electricity can be renewable or non-renewable, but electricity itself is neither renewable or non-renewable. The power plant is the location in which the energy conversions take place.

- Traditionally, large power plants have been located in sub-urban regions away from cities, as they need a vast area of land and sometimes water. All electricity produced in a power plant is alternating current (AC).

Thermal power plant

- Thermal power plant or steam power plant is a generating station which converts heat energy of fossil fuels into electrical energy. Generally **bituminous, brown or peat type coal** are used as the fuel for coal based thermal power plant.

- In a thermal power plant, coal is burnt in a big boiler which produces steam at high pressure and temperature. This steam is passed through a steam turbine which converts steam's heat energy into mechanical energy. The steam turbine acts as a prime mover and it is coupled to an alternator.

- Now alternator collects the mechanical energy from the steam turbine and convert into electrical energy. In this power plant, steam turbine sometimes acts as an auxiliary equipment's like pumps, strokes etc.

- Though thermal efficiency and overall efficiency of thermal power plant is comparatively less than other power plant like nuclear power plant but still it is very popular. The only reason behind that, is coal. It is very cheap and easily available as well. Still, now maximum energy is supplied by the coal-based thermal power plant.

TYPES OF THERMAL POWER PLANT

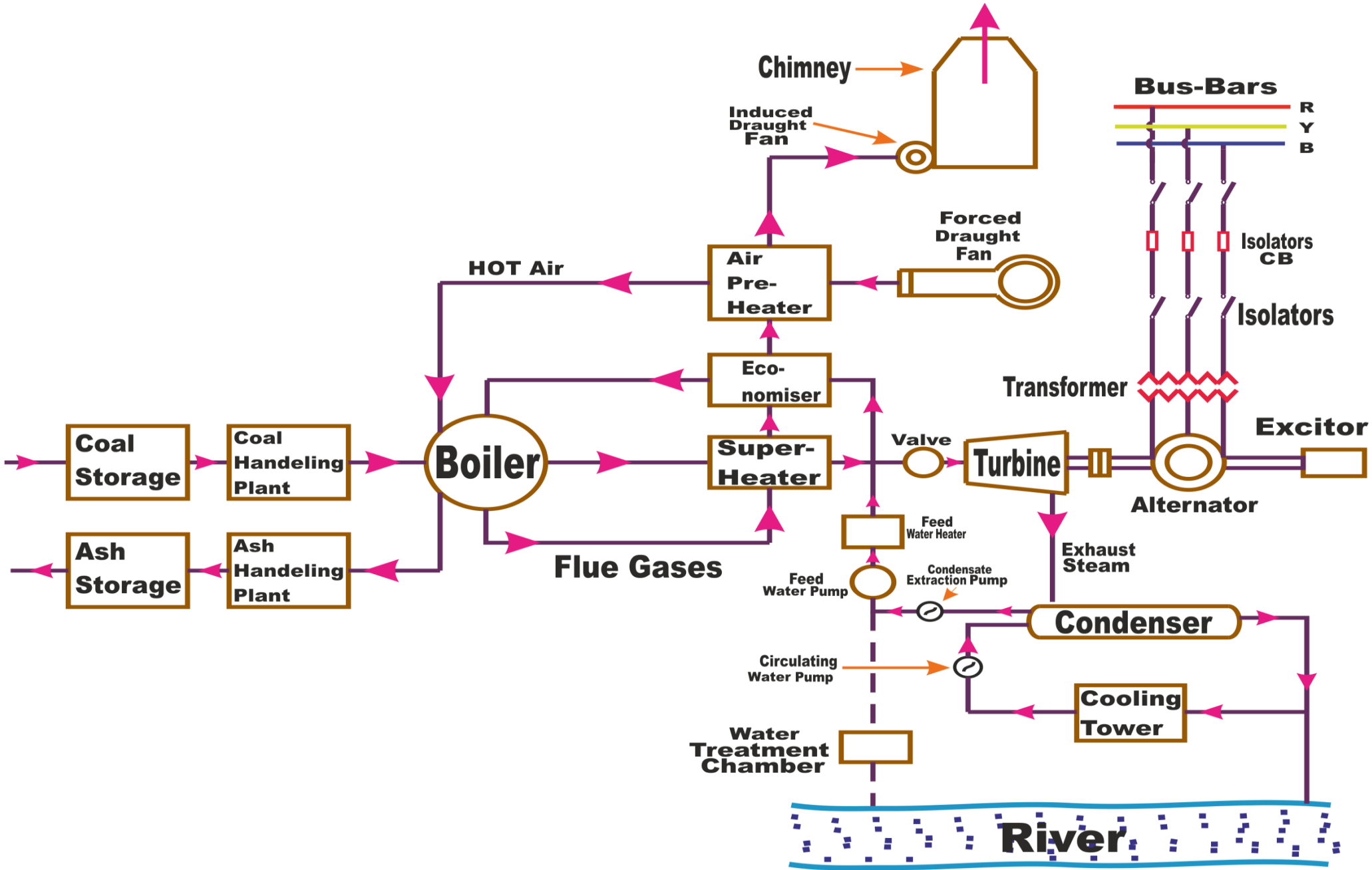
1. Condensing Type Thermal Power Plant:-

- In Condensing type thermal power plant, discharges exhaust steam is coming to a condenser which creates suction at very low pressure and it allows the expansion of steam in the turbine to a very low pressure. As a result, turbine's efficiency increases. This type of thermal power plant plays an important role where there is a question of availability of pure water. Because condensed steam water in the condenser can be re-circulated to the boiler with the help of pumps. This plant extracts **more energy from per kg of steam** and develops the greater amount of power according to the size of turbine. Condensing type plants are used to supply electrical energy all type's of consumers (domestic, industrial and commercial). Since this type of plant is very economical, so it is used both hydro power plant and thermal power plants for economical power supply all the time. The condensing type system is also used in central power plant.

2. Non-Condensing Type Thermal Power Plant:-

- In non-condensing type thermal power plant, exhaust steam is coming from turbine and steam pressure greater the atmospheric pressure. This type of thermal power plant, **a continuous supply of fresh feed water is** required. Industrial and Captive power plants are the non-condensing types. In industries, steam is used for process purpose and it is also used in the steam turbine for generation of electrical energy. This types of thermal power plant is basically small capacity power plants (nearly 10 MW).

Block Diagram of Thermal Power Plant



Different stages of power plant:

- (i) Coal and Dust Handling Arrangement

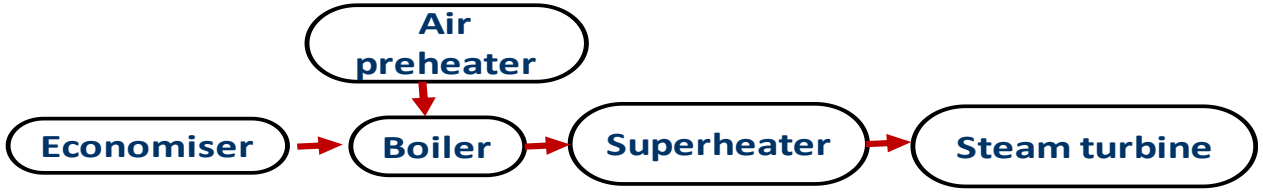


coal pulverization process

Coal combustion and ash formation

- (ii) Steam Generating Plant

Utilize flue gas heat to increase air temp (increased thermal effic and steam capacity per sq m)



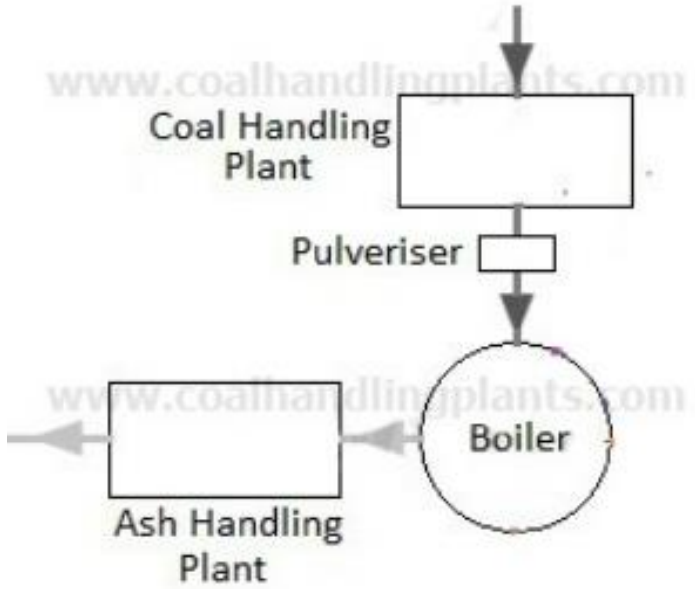
Utilize flue gas heat to increase feed water temp

Steam with high temp and pressure

dried and superheated steam (increased efficiency and prevent blade corrosion)

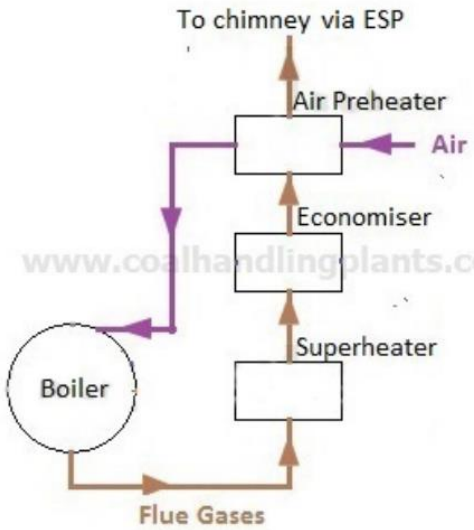
- (iii) Steam Turbine
- (iv) Alternator Generator
- (v) Feed Water
- (vi) Cooling Arrangement

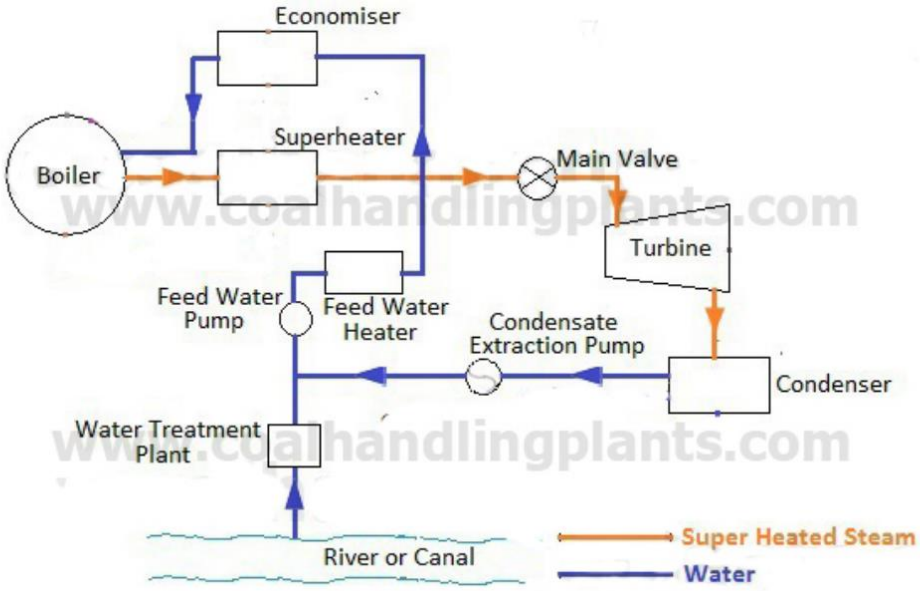
- ***1. Fuel and Ash Circuit:-***
  - The fuel (coal) is transported from supply points to power station by roads, rails or water and is stored in coal storage plant. Generally, coal is delivered into power station by rail wagon but in a case of small power plants, it may be transported by road or water. After that this coal is stored in the coal storage plant. From the coal storage plant, it is conveyed to the coal handling plant. In the coal handling plant, coal is pulverised (i.e. crushed into small pieces) to increase its surface tension to help rapid combustion without using a large amount of excess air. This pulverised coal is taken into boiler bunkers by conveyer belt. Coal is now stored into the boiler bunker and fall into the hoppers by gravity. From the hopper, required amount of coal either fall on the grate or fall into the coal spreaders. If it is fall to the coal spreaders, maximum coal combustion is done in the air and remaining portion burnt at the rear end of the grate. The grate is types of boiler, where combustion is controlled by its speed. The grate is made a move from rear end to front end with the help of spreaders or without spreaders; it is move from front to rear. The total coal combustion in the grate is controlled by its speed. After complete combustion of coal, ash is delivered to the ash storage plant by scrap conveyors for disposal. Generally, it is seen that a 100 MW power plant which operates 10% to 12% load factor may burn ***20,000 tons of coal per month and ash is produced nearly 2000 to 3000 tons per month.***



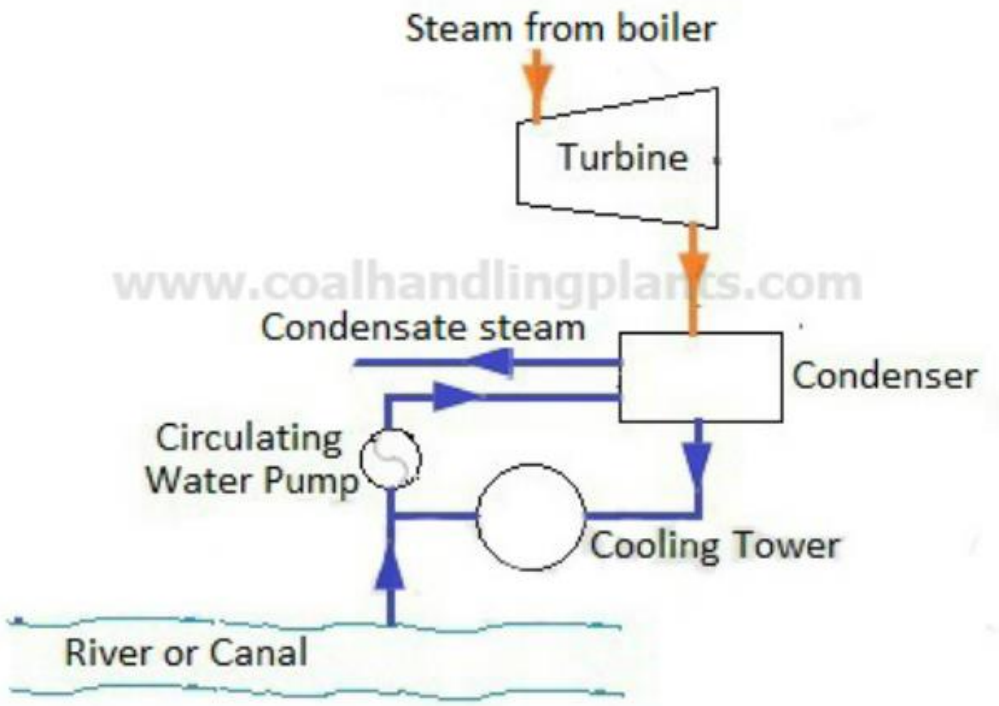
- **2. Air and Flue Gas Circuit:-**
  - The thermal power plant consists of a boiler and other auxiliary equipment which is required to utilize the air and flue gases. The air is coming from the atmosphere by a forced or induced draught fan through the air pre-heater. In the air-preheater, air is heated by the heat of flue gases which passing to the chimney. This flue gases are passing through boiler, super heater, air per heater and finally exhausted to the atmosphere through the chimney. The boiler is an essential equipment of the thermal power plant operation. In the boiler, heat is produced by coal combustion, is utilized to convert water into steam at high pressure and temperature. The steam is the wet condition in the boiler and it is converted dry and superheated condition by the flue gases when it is passed through the superheater. Superheated steam means the steam temperature is above the boiling point of water and this will provide added advantages to the turbine. This steam is increased the overall efficiency of the turbine and protect the turbine's blade from corrosion. Economiser and air preheater are such type of devices which extract the heat from flue gases on their way to a chimney and increased the temperature of the feed water. Economiser is basically a feed water heater which recovers heat from flue gases and increase the temperature of the fed water before it is supplied to the boiler. Air is also supplied for the coal burning. Air preheater also extracts the heat from flue gases when it is passed in it. Air preheater increases the heat of air as well as improves the overall efficiency of the turbine. It also ***increases the steam capacity per square meter of boiler surface***. Now this dry and superheated steam is fed to the blades of steam turbine through the main valve. Here steam energy is converted to the mechanical energy.

- **3. Feed water and steam circuit:-**
  - Condensed steam is coming out of the turbine and the condensate is extracted from the condenser by the condensate extraction pump. Exhaust steam is passed through the low-pressure feed water heater where its temperature is raised by the bled steam. The feed water is now pumped by deaerator to high-pressure feed water heater where this feed water is heated by the heat from bled steam extracted at suitable point of steam turbine. Deaerator is to reduce dissolved oxygen content in the feed water. Feed water is pumped into the boiler and it is passing through the economiser where it is heated by the heat of flue gases. This will increase the overall efficiency. Some steam and water are lost when it is passed through the different component of the system. Turbine is directly coupled to the alternator which converts the mechanical energy of turbine into electrical energy and delivers the electrical output to the bus bar.





- 4. Cooling water circuit:-
  - Cooling water is supplied from a natural source of supply such as river, channel, sea, etc. This water is circulated through the condenser for condensing the steam. It will increase the overall efficiency of the plant. The circulating water absorbs heat from the exhaust steam and becomes hot. This hot water is coming out from the condenser and discharged at a suitable position like a lake, river etc. To ensure the availability of cold water throughout the year, a cooling tower is used. During the scarcity of water, hot water of condenser is passed to the cooling tower where it is cooled. The circulation of cooling water to the condenser is to maintain low pressure in the condenser.



SITE FOR THERMAL POWER STATIONS

- **Cost and Type of Land** : Cheap (rural areas) and the land bearing capacity of should be adequate to install heavy machinery.
- **Availability of Water :** A large amount of water is required for condensers 560 × 10^3 kg of water is required for every 1 tonne of coal burnt. ----bank of a river or near a canal
- **Supply of Fuel :** Near the coal mines so that transportation cost of fuel is minimum.
- **Nearness to Load Centre :** In order to reduce the transmission cost, the plant should be located near the center of the load.
- **Distance from Populated Area :** The plant should be located at a considerable distance from the populated area to minimize pollution.
- **Ample Space :** Future expansion
- **Disposal of Ash :** Facility for the disposal of ash.

ADVANTAGES OF THERMAL POWER PLANT:

- The fuel (i.e. coal) is used is quite cheap.
- Less initial cost as compared to the other generating stations of the same capacity.
- It requires less space as compared to the hydro-electric power station.
- The cost of generation is lesser than the Diesel power station.
- According to the demand, the load can be changed frequently without any difficulty.
- Thermal Power plant can be installed anywhere irrespective of the availability of fuels. Fuel can be transferred to the site of the plant by rail, road etc.
- This type of plants is installed near load centre.
- Thermal Power plant can be run with overload condition (around 25%).

DISADVANTAGES OF THERMAL POWER PLANT:

- It pollutes the atmosphere due to the production of a large amount of smoke.
- Maintenance cost and operating cost is high.
- A Huge amount of water is required.
- Running cost is high as compared to the other power plant.
- Coal handling and ash disposal is quite difficult.

Hydroelectric power plant

- Hydroelectric power is developed by utilizing the ***potential energy of water***.

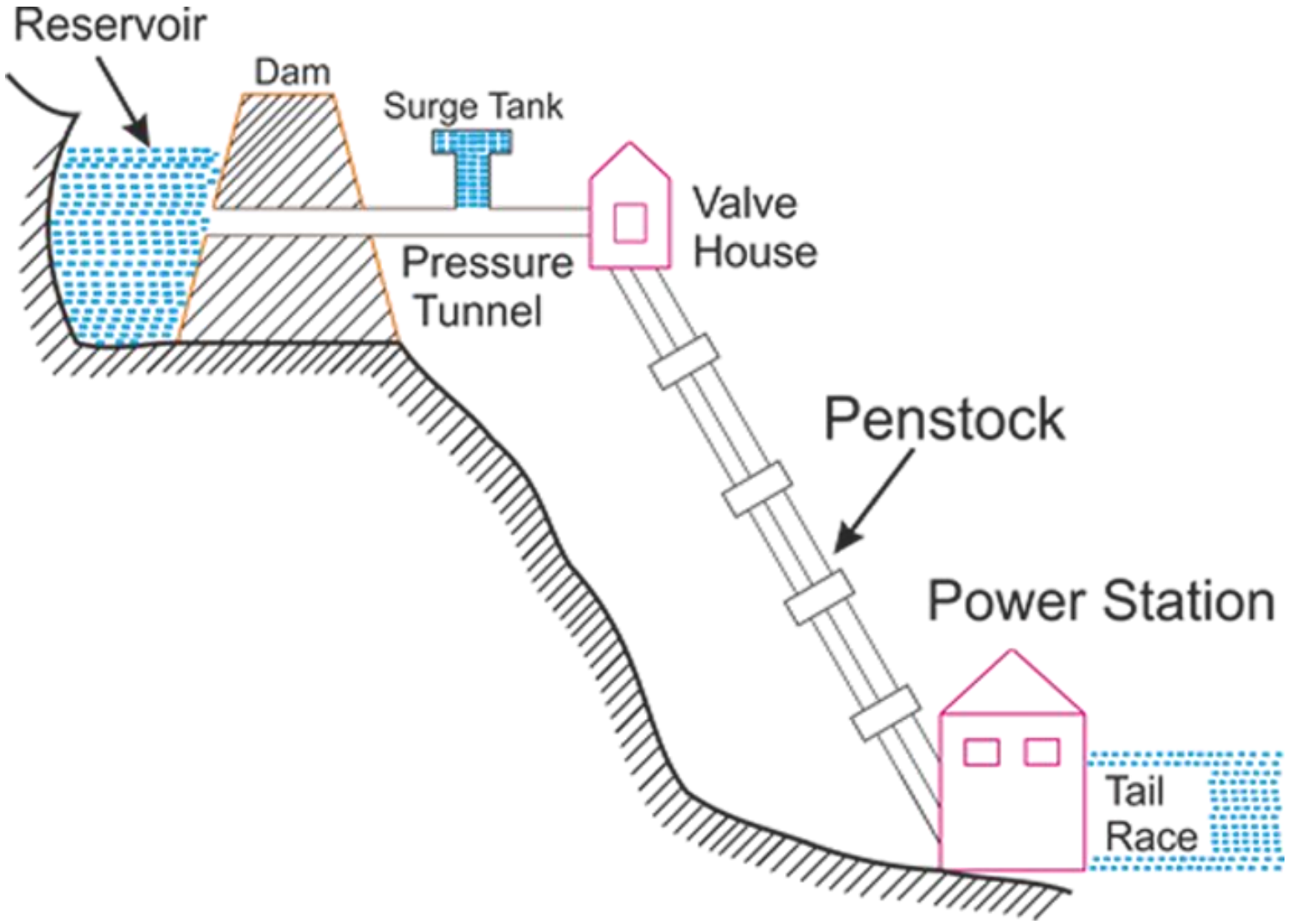
- In hydroelectric power plant, water is stored in a dam called hydroelectric dam which is located upper level from the ground especially any hilly areas. Water head is created by construction the dam across any river or lake. This type of water head store huge potential energy.

- The water fall into water turbine and the ***potential energy of water is converted into kinetic energy***.

- ***Kinetic energy is converted into mechanical energy*** at the turbine shaft. A hydroelectric generator or alternator is coupled with turbine shaft to convert ***mechanical energy into electrical energy***. The power P is developed-

$$P = \eta \rho Q g h$$

Block diagram of hydroelectric power plant



Working Principle of hydroelectric power plant

Hydroelectric power plant is becoming very popular nowadays to full fill rapid increasing demand of electric power day by day. Every country is trying to develop more Hydro Electric Power Station to full fill their demand for electricity. In other hand fossils, fuels (i.e. coal, oil, and gas) are limited stock in the world and these fuels are expensive. So hydroelectricity may be a good alternative electrical source. So in a single word we can say, a generating Station which utilizes the potential energy of high-level water for the generating of electrical energy is known as hydropower plant or hydroelectric power plant.

- conventional hydroelectric facilities- rely on a head difference created by man-made dams and obstructions. Two types of systems that are considered conventional are hydroelectric dams and tidal dam.
- unconventional hydroelectric facilities- rely on hydroelectric discharge or on a small head differential. Some examples of unconventional hydropower facilities are low head hydro, run-of-the-river systems, instream hydro, and kinetic tidal.

- Working principle of hydroelectric power plant depends on the conversion of hydraulic energy into electrical energy.

- Hydroelectric power station needs huge amount of water at sufficient head all the time. So a hydroelectric dam is constructed across the river or lake.

- An artificial storage reservoir where water is stored, is placed back side of the dam. This reservoir creates sufficient water head.

- A pressure tunnel is placed in between the reservoir to valve house and water is coming from reservoir to penstock via this tunnel. An automatic controlling sluice valve is placed in valve house and it controls water flow to the power station and the letter cuts off supply of water in case the penstock bursts. Penstock is a huge steel pipe in which water is taken from valve house to turbine. A surge tank is also provided just before the valve house for better regulation of water pressure in the system. Now water turbine converts hydraulic energy into mechanical energy and an alternator which is couple to the water turbine converts this mechanical energy into electrical energy.

SITE SELECTION FOR HYDROELECTRIC POWER PLANT

1.**Water Availability:**

Main fuel of this plant is water. So, such plant should be located nearer to river, canal etc. where sufficient water is available all the time.

2.**Water Storage:**

Storage of water in a suitable reservoir or dam has to be placed by a careful geological study of the area to get the maximum advantage of that water. Dam should be located across the river to get continuous water supply throughout the year specially in a dry season. The storage capacity of dam can be determined by hydrograph or mass curve or using analytical method. Adequate facilities of erection a dam and storage of water are two important matters for site selection of hydro electric power plant.

3.**Water Head:**

It is an important point for site selection of hydroelectric power plant. Water head is directly related to the cost of generation of electric power. If effective head is increased, water storage has to be reduced as well as capital cost of the plant is reduced.

4.**Distance from the load center:**

Since it is located away from the load center, more transmission line is required to supply the power. To avoid the line loss and economical power supply, distance of such plant should need more attention.

5.**Transportation Facilities:**

Good transportation facilities must be available to any hydro electric power plant, so that necessary equipment should be reached easily.

6.**Availability of land:**

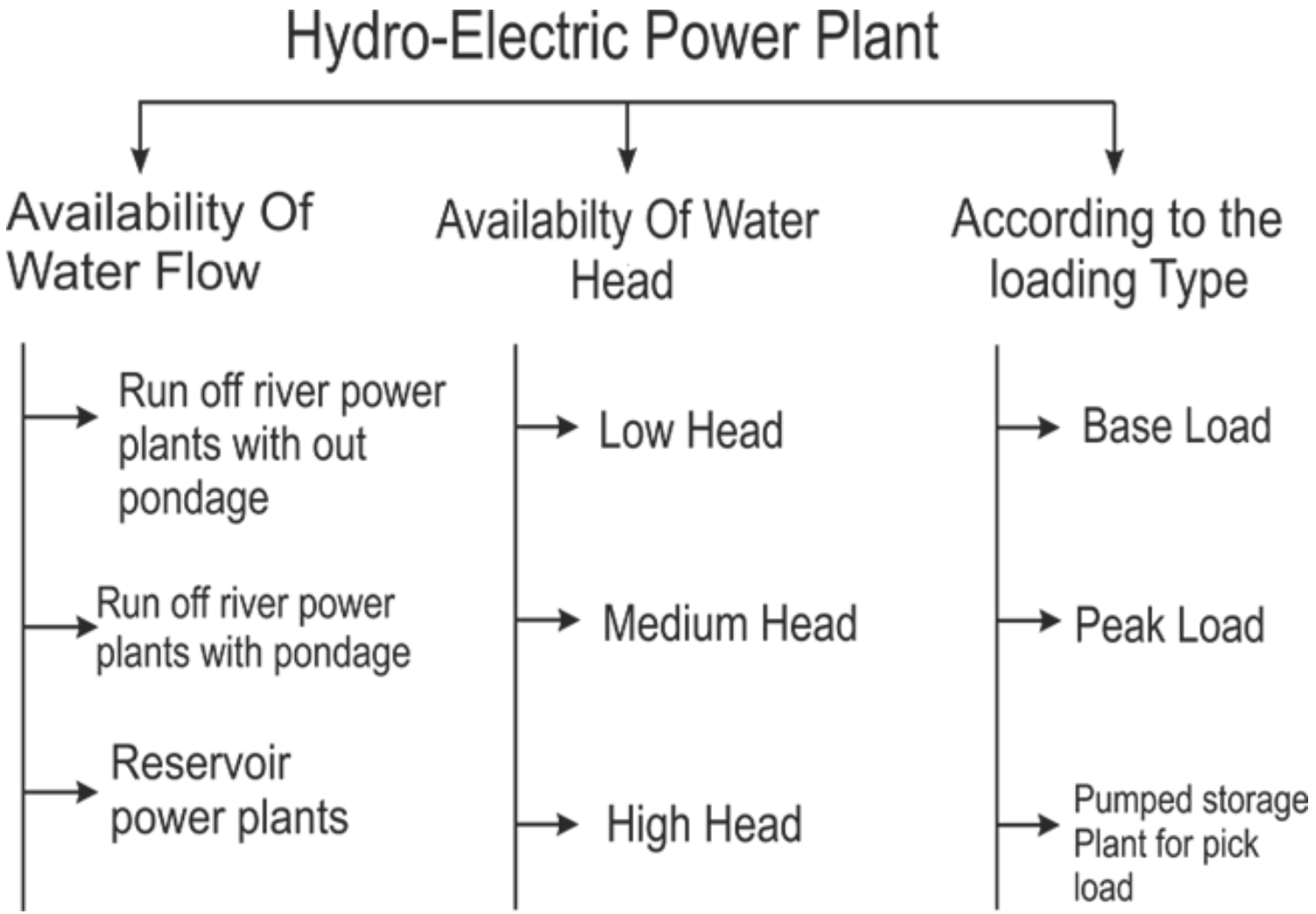
Hydro electric power plant needs enough space. It should be kept in mind that land cost must be cheap.

Advantages

- Since water is the main source of energy, so no fossil fuels are required.
- This plant is neat and clean and no smoke or as disposal is required.
- It is the cheapest operating and maintenance cost as compared to the other power plants because water is freely available in the world.
- It is very reliable, robust and has a longer life app rocks 45 to 60 years.
- This plant can start instantly.
- It can start hydroelectric power with fluctuating load demand.
- The efficiency does not fall at the age of this plant.
- There is no standby loss in this plant.
- At the initial time of construction highly skilled engineers are required and after that only few experience persons can run the plant.
- This plant also serves to help in irrigation and Flood control etc.
- Since this plants are located remote area so land is available and competitively cheaper rates.

Disadvantages

- Such plant requires large area
- High construction cost is required due to construction of dam.
- When experience skilled engineers are required to build this plant
- Scenes such plant is located as from the load areas, long transmission line is required to transmit this hydroelectric power.
- It doesn't supply constant hydroelectricity due to the availability of water. In transition, power supply is most affected.

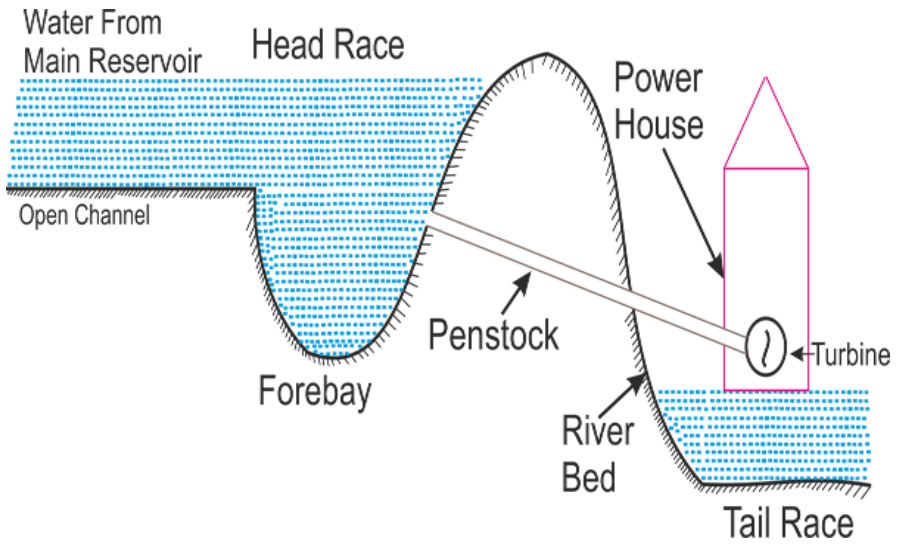

Hydro-Electric Power Plant
Availability Of Water Flow
Availabilty Of Water Head
According to the loading Type
Run off river power plants with out pondage
Run off river power plants with pondage
Reservoir power plants
Low Head
Medium Head
High Head
Base Load
Peak Load
Pumped storage Plant for pick load

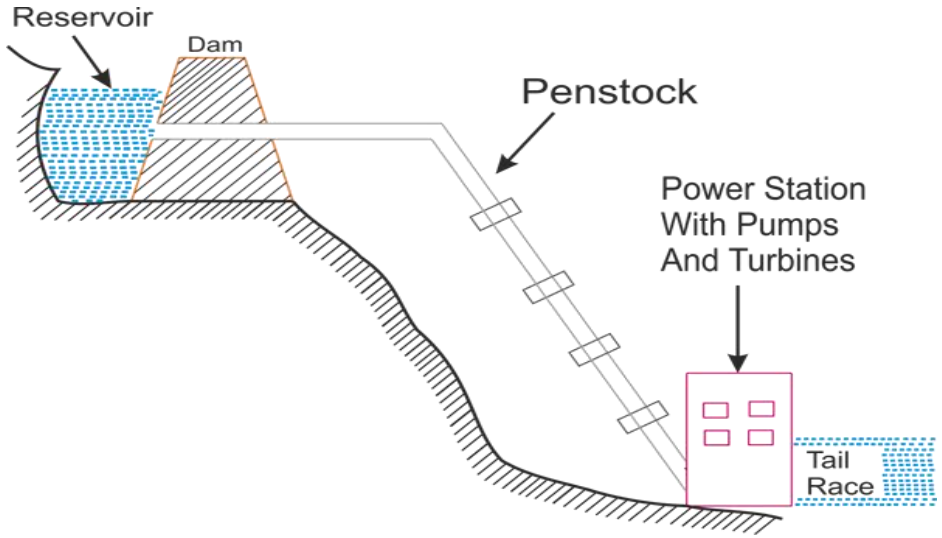
- **Run off river power plants without pondage**
  - Water is not available all the time.
  - Not suitable for constant steady load. There is no pondage or storage facility available in such type of power plant. Plant is placed in such a area, where water is coming directly from the river or pond. This type of hydroelectric power plant is called run off power plant without pondage. Plant produces hydro electricity only when water is available.

- **Run off river power plants with pondage**
  - This type of plant is used to increase the capacity of pond. The pond is used as a storage water of hydro electric power plant. Increased the pond size means more water is available in the plant,
  - Used fluctuating load period depending on the size of pondage.

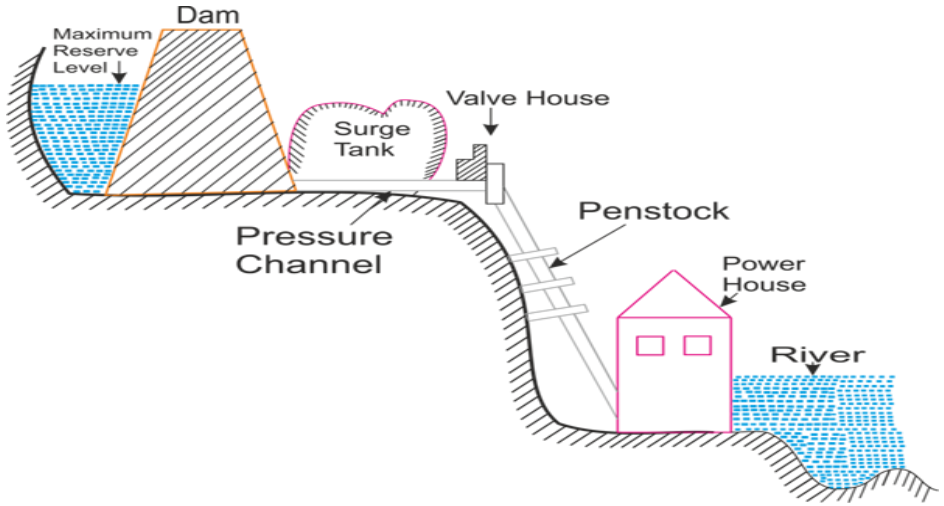
- **Reservoir power plants**
  - Water is stored behind the dam and water is available throughout the year even in dry season.
  - Very efficient and it is used both base and peak load period

- **a. Low head hydro electric power plant**
  - water head height is below 30 meters
  - dam construction is essential.
  - No surge tank is required for this plant, dam itself discharge the surplus water from the river. large diameter and low length pipe is used for this plant





- **b. Medium head hydro electric power plant**
  - water head height is above 30 meters to 300 meters
  - A forebay is used for medium head hydro electric power plant. Which act as a surge tank.
  - Forebay is tapped with the river and water is led to the turbine via penstock.



- **c. High head hydro electric power plant**
  - head of this power plant is more than 300 meters.
  - For high head more than 500 meters, Pelton wheel turbine is used and for lower head Francis turbine is useful.

- **a. Base load hydro electric power plant**
  - Base load plant is suitable for constant load.
  - load factor of this plant is high
  - Run off river plants without pondage and reservoir plants are used as base load plants.

- **b. Peak load hydro electric power plant**
  - This plant is suitable for peak load curve of power system.
  - Run off river plants with pondage and Reservoir plants can be used as peak load plants.

- **c. Pumped storage hydro electric power plant for the peak load**
  - Two types of water pond is used, called upper head water pond and tail water pond.
  - Two water ponds are connected each other by a penstock.
  - During the off load period, surplus energy of this plant is utilized to pumping the lower head pond water to upper head pond water.
  - This extra water is used to generate energy at peek load periods.
  - By doing this arrangement, same water is used again and again.

India's Pumped-Storage Requirement

2026-27 NEP benchmark
~7.45 GW / 47 GWh

2031-32 NEP benchmark
~26.69 GW / 175 GWh

Estimated PSP investment by 2031-32: ~₹1.29 lakh crore

- These planning benchmarks reflect the storage needed to support India's rapidly growing solar and wind fleet.

Pumped Storage Pipeline: 2026 Update

Under construction
15.87 GW

Energy storage
95.22 GWh

100 GW of Hydro Pumped Storage Projects by 2035-36

- The newer 100-GW roadmap shows that India's storage ambition is moving beyond the older NEP baseline.

Why Does India Need So Much Pumped Hydro?

Solar / Wind ↑ → **Variable generation & evening ramp** → **Need flexibility & storage**

Future grid = Solar/Wind + BESS + Pumped Hydro + Flexible Generation + Transmission

- Forecasting identifies when surplus or deficit is likely.
- Storage shifts electricity from low-value/surplus periods to high-demand periods.

Battery Storage vs Pumped Hydro

Parameter	BESS	Pumped Hydro
Storage principle	Electrochemical	Gravitational
Response	Extremely fast	Fast
Typical strength	Short-duration / flexible	Bulk / long-duration
Lifetime	Lower	Very long
Siting	Highly flexible	Geography dependent
Construction	Generally faster	Generally slower
Grid services	Excellent	Excellent

BESS and PSP are complementary—not necessarily competing technologies.

Modern Hydropower Is More Than Bigger Dams

Closed-loop / off-river PSP
Reuse water between two reservoirs

Variable-speed pump-turbines
Higher operating flexibility

Digital governors & controls
Faster, smarter grid response

AI / predictive maintenance
Detect faults before failure

Hydrological forecasting
Better inflow & reservoir operation

Digital twins / remote monitoring
Condition-based operation

Advanced CFD turbine design
Higher efficiency, lower cavitation

Hybrid RE + hydro operation
Solar/wind firming & balancing

Case Study: Subansiri Lower HEP

Capacity
2,000 MW

Units
8 × 250 MW

Dam
116 m

Design energy
~7,422 MU/yr

- Run-of-river scheme with small pondage.
- Uses eight head-race tunnels.
- Incorporates a flood cushion of about 442 million m^3.
- First 250-MW unit entered commercial operation in late 2025.
- Useful teaching example of power generation combined with water-management functions.

India's Hydro Outlook to 2031-32 and Beyond

Large Hydro
~62.18 GW
NEP 2031-32 projection

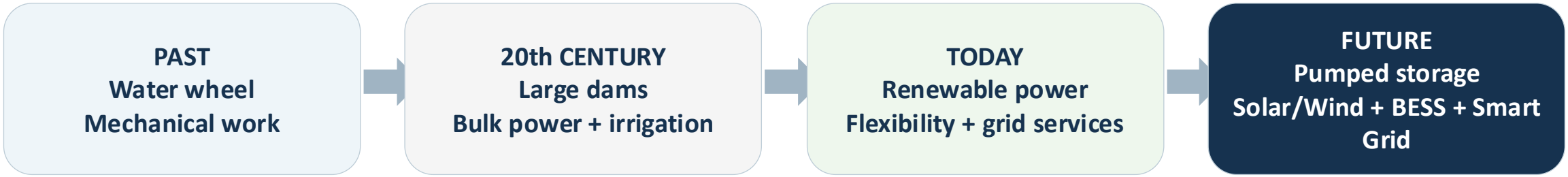
Pumped Storage
~26.69 GW
NEP 2031-32 projection

Newer PSP roadmap
100 GW
by 2035-36

Hydro's strategic value is shifting from only 'energy generation' to 'flexibility + storage + grid services'.

- Recent planning indicates a large construction pipeline for both conventional hydro and pumped storage.

Evolution of Hydropower: Past → Future



Nuclear power plant

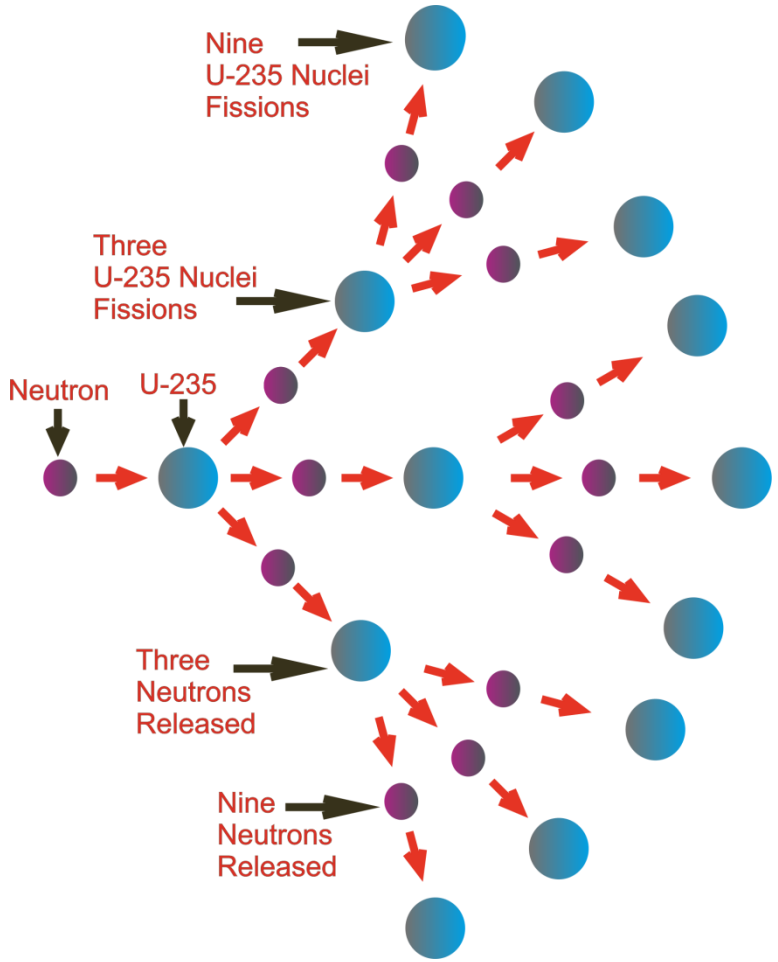
- It can produce a huge amount of electrical energy from relatively small amount of nuclear fuel as compared to other conventional types of power stations.
- It is seen that complete fission of 1 kg nuclear fuel like uranium-235 would produce energy equivalent to 4000 tonnes of high-grade coal.
- Although nuclear fuels (like uranium, thorium) are difficult to recover and these are expensive too but nuclear fuel cost is considerably less than the conventional steam power station.
- Fuel cost of nuclear power plant is generally 40% cheaper than a conventional steam power station.
- Nuclear energy is converted to electrical energy is known as a ***nuclear power plant***.

NUCLEAR CHAIN REACTION PROCESS:

- **Uranium-235** is natural and it will give higher fission percentage too.
- Some other fissionable materials (U-233, PU-239) are present in the world.
- U-235 is the most common isotope to use for a nuclear chain reaction.
- It produces comparatively higher amount of energy and heat from the other fissionable materials.

- $^{235}_{92}U + {}^{1}_{0}n = {}^{139}_{56}Ba + {}^{94}_{36}Kr + 3\ {}^{1}_{0}n$

- In the above reaction, it is shown that a heavy nucleus uranium-235 strikes one neutron and it produce alternative three neutrons. These three neutrons again strikes U-235 which produce nine neutrons and striking nine neutrons produce another twenty-seven neutrons and so on, showing in the picture
- A single fission reaction generates a large amount of energy, so the complete nuclear chain reaction process will generate many times greater energy.
- It is seen that one fission reaction of U-235 release nearly 200 MeV of energy.
  200MeV energy: 200 X 1.6 X 10^{-13} = 3.2 X 10^{-11}joules (or watt-seconds)



ADVANTAGES OF NUCLEAR POWER PLANTS

- There are lots of **advantages of nuclear power plant** as compared to other power plants.
- Since the requirement of fuel is very small, so the cost of fuel transportation, storage etc. is small.
- Nuclear power plant needs less space as compared to any other power station of the same size. Example: A 100 MW nuclear power station needs 38 - 40 acres of land whereas the same capacity coal based thermal power plant needs 120-130 acres of land.
- This type of power plant is very economical to produce large electric power.
- Nuclear power plant can be located near load center because bulk amount of fuel (like water, coal) is not required.
- Nuclear power is most economical to generate large capacities of power like 100 MVA or more. It produces huge amount of energy in every nuclear fission process.
- Using a small amount of fuel, this plant produces large electrical energy.
- This plant is very reliable in operation.
- Since, the large number of nuclear fuel is available in this world. So, a nuclear power plant can generate electrical energy thousands of years continuously.
- Nuclear Power Plant is very neat and clean as compared to a steam power plant.
- The operating cost is low at this power plant but it is not affected for higher load demand. Nuclear power plant always operates a base load plant and load factor will not be less than 0.8.

DISADVANTAGES OF NUCLEAR POWER PLANTS

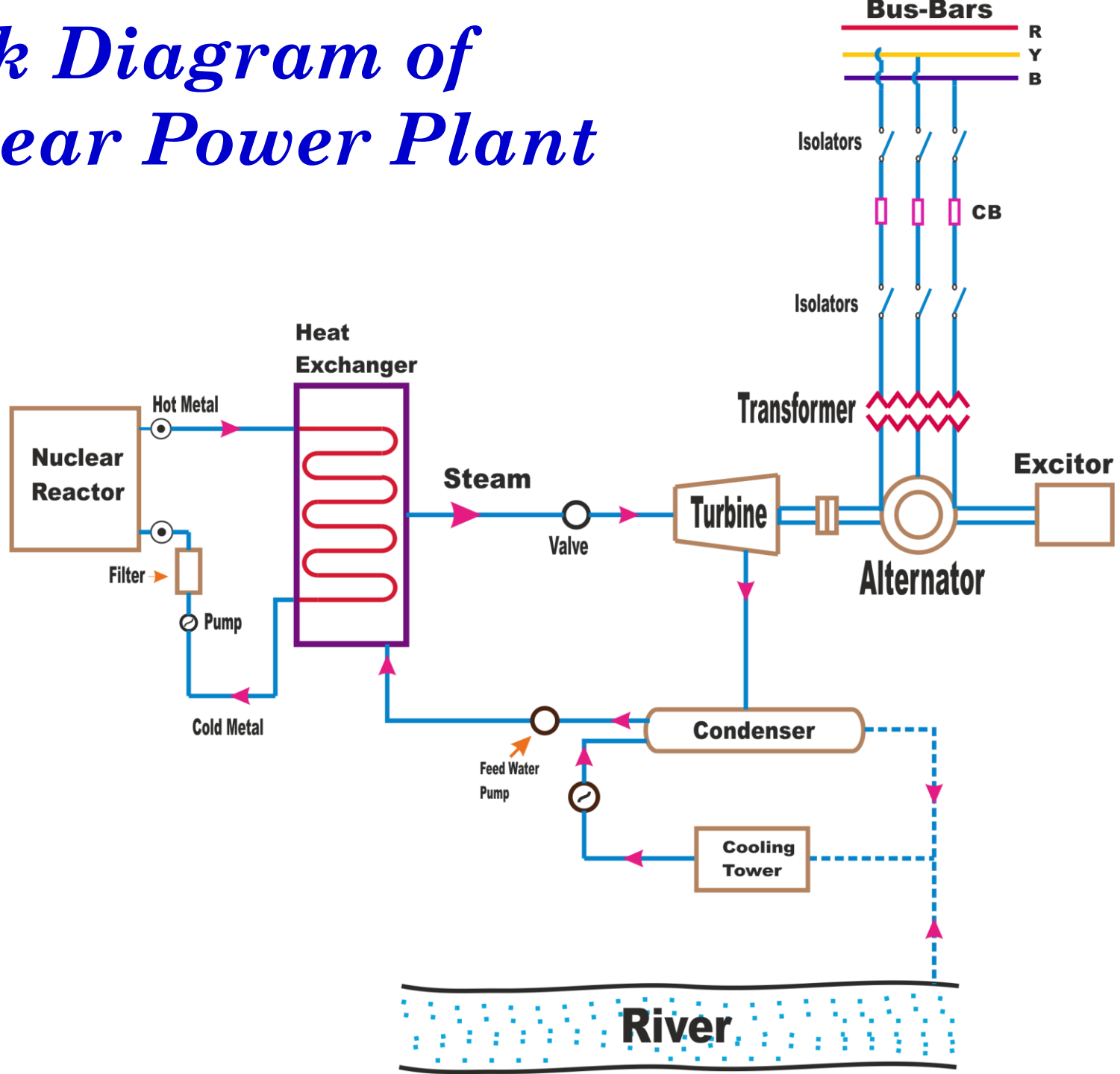
- Initial installation cost is very high as compared to the other power station.
- Nuclear fuel is very much expensive and it is difficult to recover.
- Capital cost is higher in respect of other power station.
- Good technical knowledge is required to operate such type plant. So, salary bill and other maintenance cost will be higher to operate such of a plant.
- There is a chance to spread of radioactive pollution from this type of plant.
- Nuclear Reactor does not response efficiently with the fluctuating load demand. So, it is not suited for varying the load.
- Cooling water requirement is twice than a coal based steam power plant.

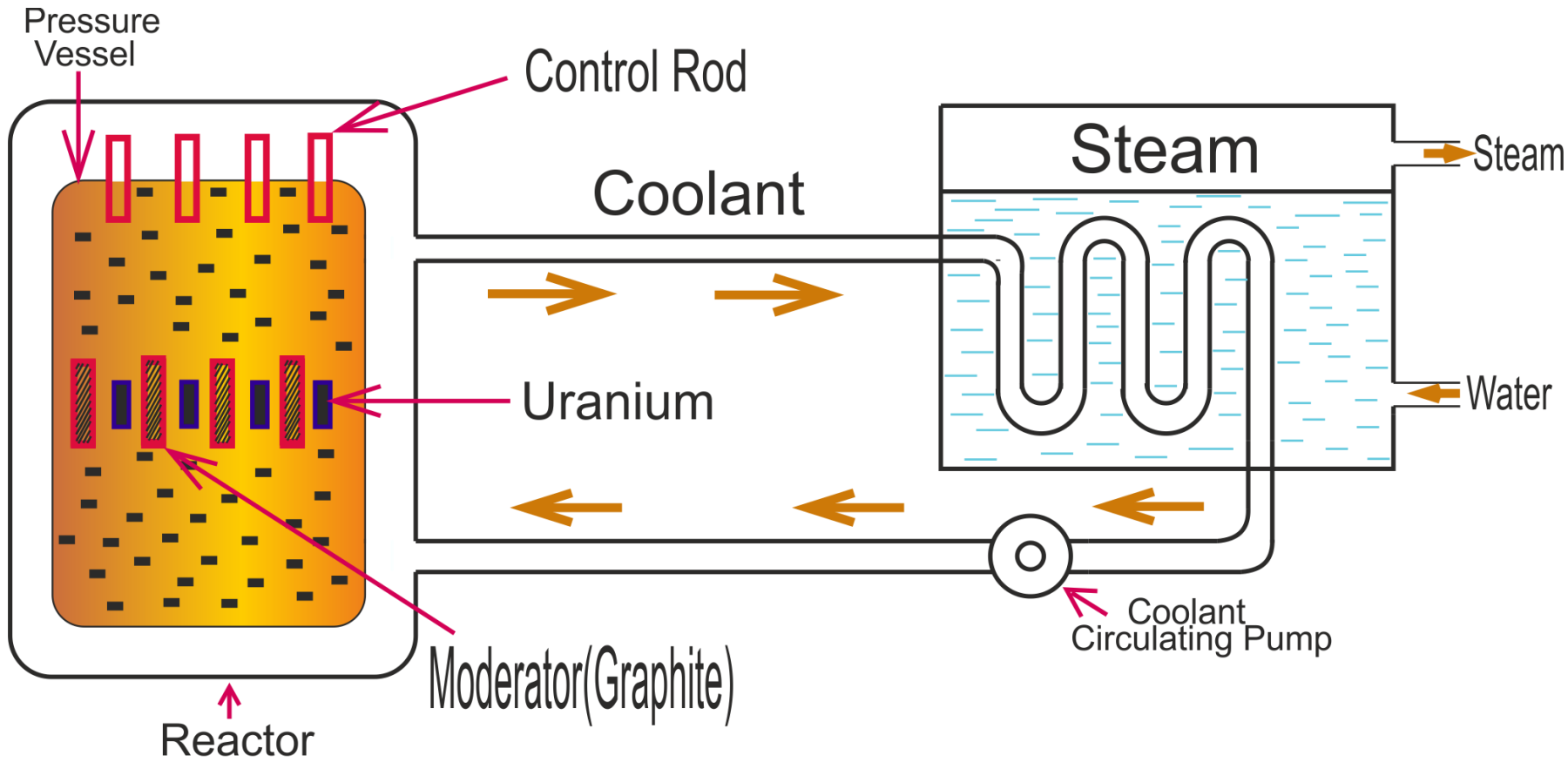
Working Principle of nuclear power plant

1. **Nuclear Reactor**
2. **Heat Exchanger**
3. **Steam Turbine**
4. **Alternator**

Nuclear reactor is used to produce heat and heat exchanger performs to convert water into steam by using the heat generated in nuclear reactor. This steam is fed into steam turbine and condensed in condenser. Now steam turbine is turn to run an electric generator or alternator which is coupled to steam turbine and thereby producing electric energy.

Block Diagram of Nuclear Power Plant





Nuclear Reactor

1. Nuclear Reactor:-

- ***Nuclear reactor***, where nuclear fuel is subjected to nuclear fission.
- By the braking of uranium atom, tremendous amount of heat energy and radiation is formed in the reactor and the chain reaction is continuously running until it is controlled by a reactor control chain reaction.
- A large amount of fission neutrons are removed in this process, only small amount of fission uranium is used to generate the electrical power.
- The nuclear reactor is cylindrical type shape.
- Main body of reactor is enclosed by ***reactor core, reflector and thermal shielding***. It prevent reactor wall from getting heated. It is also used to protect alpha (α), bita (β) , gama (γ) rays and neutrons which are bounce back at the time of fission within the reactor.
- Mainly Nuclear reactor consists, some fuel rods of uranium, moderator and control rods.
- Fuel rods are made of the fission materials and released large number of energy at the time of bombarding with slow moving neutrons.
- Moderator consists full of graphite which is enclosed by the fuel rods. Moderator maintains the chain reaction by releasing the neutrons in a suitable manner before they mixed with the fissile materials.
- Control rods are made of boron-10 and cadmium or hafnium which is a highly neutron absorber and it is inserted into the nuclear reactor. When control rods are push down into the reactor core, it absorbs most of fission neutrons and power of the reactor is reduced. But when it is pulling out from the reactor, it releases the fission neutrons and power is increased. Real practice, this arrangement depends upon according to the requirement of load.
- A coolant, basically sodium metal is used to reduce the heat produce in the reactor and it carries the heat to the heat exchanger.

2. Heat Exchanger:-

Coolant is used to raise the heat of the heat exchanger which is utilised in raising the steam. After that, it goes back to the reactor.

3. Steam Turbine:-

Steam is coming from the heat exchanger to fed into the steam turbine through the valve. After that the steam is exhausted to the condenser. This condensed steam is fed to the heat exchanger through feed water pump.

4. Alternator:-

Steam turbine is coupled to an alternator which converts mechanical energy to electrical energy. The output of alternator produces electrical energy to bus bars via major electrical apparatus like transformer, circuit breakers, isolators etc.

India's Three-Stage Nuclear Programme

Closed fuel-cycle strategy: use limited uranium to unlock India's much larger thorium resource

STAGE 1 • PHWR

Natural uranium (UO_2)
≈0.7% U-235 + 99.3% U-238

U-235 → fission → electricity
U-238 + n → Pu-239

Spent fuel is reprocessed; plutonium becomes fuel for Stage 2.

STAGE 2 • FBR

Pu-239-based fuel + fertile U-238

Fast neutrons → electricity + breeding of additional fissile material

Thorium-232 can be loaded in blanket / low-power zones to breed U-233 for Stage 3.

STAGE 3 • THORIUM

Th-232 is fertile, not directly fissile.

Th-232 + n → Th-233 → Pa-233 → U-233

U-233 is fissile → fission → electricity

Goal: self-sustaining Th-232 / U-233 fuel cycle.

Core logic **Natural U → Pu-239 → U-233 → long-term use of abundant thorium**

FISSILE: U-235 • Pu-239 • U-233

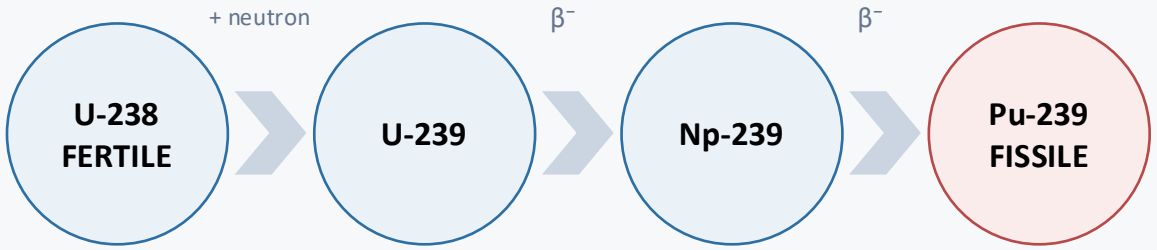
FERTILE: U-238 → Pu-239 | Th-232 → U-233

Sources: BARC — Indian Nuclear Power Program; BARC — Thorium Fuel Cycle; BARC AHWR programme (accessed Sep 2026).

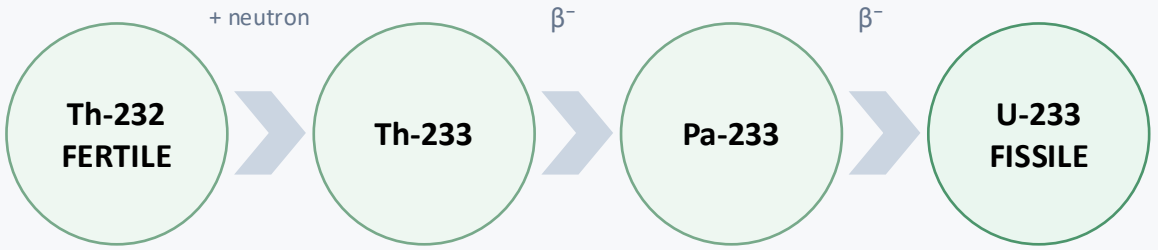
How the Fuel “Breeding” Links the Three Stages

The programme is sequential because thorium needs fissile material and neutrons before it can become a practical fuel cycle

A. Uranium → Plutonium (build fissile inventory)



B. Thorium → Uranium-233 (unlock Stage 3)



Stage 2 milestone: 500 MWe PFBR, Kalpakkam

Prototype Fast Breeder Reactor
- Fast-neutron spectrum; no moderator
- Sodium-cooled breeder technology
- Strategic bridge from PHWR plutonium to large-scale thorium utilisation

Why a CLOSED fuel cycle?

Spent fuel is treated as a resource:
PHWR spent fuel → recover Pu → FBR fuel
FBR / thorium irradiation → U-233 → Stage 3

Purpose: multiply the usable energy from domestic nuclear resources and reduce dependence on fresh uranium.

Key idea: Fertile material does not “turn back into natural uranium” — Th-232 is converted into fissile U-233.

Sources: BARC — Indian Nuclear Power Program; BARC newsletters on closed fuel cycle / three-stage programme.

Where SMRs Fit in India's Nuclear Future

SMRs complement the three-stage fuel strategy; they are a deployment pathway, not a fourth stage

SMR

Small Modular Reactor

Typically ≤300 MWe/module

Smaller unit size + modular / standardized construction

Potential uses:
- Captive industrial power
- Retiring coal-site repurposing
- Remote/off-grid applications
- Process heat / hydrogen

India's indigenous demonstration portfolio

BSMR-200 — 220 MWe

Power generation

SMR-55 — 55 MWe

Smaller power applications

HTGCR — ≤5 MWth

Process heat / H_2

Nuclear Energy Mission: ₹20,000 crore for SMR R&D/deployment • ≥5 indigenous SMRs by 2033

How SMR relates to the 3-stage programme

THREE-STAGE PROGRAMME

answers:

"Which nuclear fuel cycle should India develop for long-term resource security?"

SMR PROGRAMME

answers:

"How can nuclear capacity be deployed in smaller, modular, industrial and brownfield applications?"

SMR ≠ Stage 4

Fuel-cycle strategy and reactor deployment strategy are different dimensions.

Sources: PIB/DAE, 5 Aug 2026 — BSMR-200 220 MWe, SMR-55 55 MWe, HTGCR ≤5 MWth; ≥5 indigenous SMRs by 2033; ₹20,000 crore NEM provision.

Thermodynamics

- Thermodynamics, science of the relationship between heat, work, temperature, and energy.
- At present, fossil fuel is still the world's predominant energy source. Burning of fossil fuels generates only thermal energy.
- To convert thermal energy into another form of energy a heat engine must be used.
- Thermodynamic equilibrium -A state of a physical system in which the system is in **mechanical, chemical, and thermal equilibrium** and in which there is no tendency for spontaneous change.

Thermodynamic System

- A thermodynamic system refers to that part of the universe in which observations are made, and the remaining universe constitutes the surroundings.

- The surroundings contain everything other than the system. The system and the surroundings together make up the universe.

- Boundary- separate system & surrounding; fix, movable and imaginery

- Thermodynamic systems are classified as :
    - Open systems (turbine, pump, compressor)
    - Closed systems (piston cylinder without valves)
    - Isolated Systems (universe, hot coffee in a perfectly insulated thermos)

State ,Path and Process in Thermodynamics

- A state is condition of system and is specified by its properties. At a given state all the properties of a system have definite values.

- Change of state is known as process.
- Change of state of a system is the consequence of any operation in which properties will change. The series of states through which system passes during a change of state is called the part of the process.

- A thermodynamic process (state 1 → state 2) is said to be reversible if the process can be turned back to such that both the system and the surroundings return to their original states, with no other change anywhere else in the universe.
- A process that eventually returns a system to its initial state is called a cyclic process. At the conclusion of a cycle, all the properties have the same value they had at the beginning.

Thermodynamic cycles

- ***A process that eventually returns a system to its initial state is called a cyclic process.***
- ***At the conclusion of a cycle, all the properties have the same value they had at the beginning.***
- ***Thermodynamic cycle consists of a series of thermodynamic processes transferring heat and work, while varying pressure, temperature, and other state variables, eventually returning a system to its initial state.***
- ***The thermodynamic cycles can be divided into two primary classes:***

  - ***Power cycles-***
    - ***Power cycles are cycles which convert some heat input into a mechanical work output.***
    - ***Thermodynamic power cycles are the basis for the operation of heat engines, which run the vast majority of motor vehicles and generate most of the world's electric power.***
  - ***Heat pump cycles-***
    - ***Heat pump cycles transfer heat from low to high temperatures using mechanical work input.***
    - ***There is no difference between thermodynamics of refrigerators and heat pumps. Both work by moving heat from a cold space to a warm space.***

Thermodynamic cycles

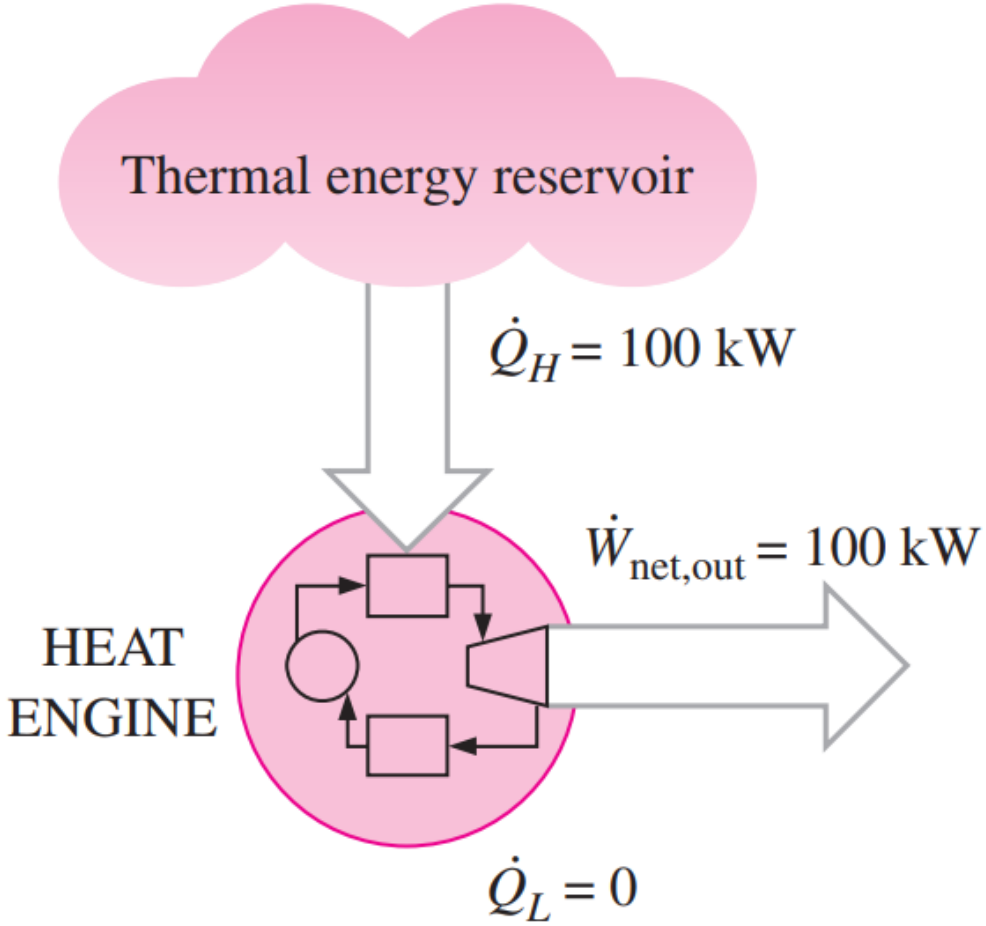
- Classification of thermodynamic cycles according to their thermodynamic processes.
- In practice, simple idealized thermodynamic cycles are usually made out of four thermodynamic processes.
- The following processes usually constitute thermodynamic cycles:
    - Adiabatic process-
        - No heat transfer into or out of the system (Q = 0).
        - The system can be considered to be perfectly insulated.
        - In an adiabatic process, energy is transferred only as work.(very rapid process)
    - Isothermal process-
        - Temperature of the system remains constant (T = const).
        - The heat transfer into or out of the system typically must happen at such a slow rate in order to continually adjust to the temperature of the reservoir through heat exchange
    - Isobaric process-
        - pressure of the system remains constant (p = const).
        - The heat transfer into or out of the system does work, but also changes the internal energy of the system.
    - Isochoric process-
        - volume of the closed system remains constant (V = const).
        - the heat transfer into or out of the system does not depend on pΔV work, but only changes the internal energy (the temperature) of the system.

Law of Thermodynamics

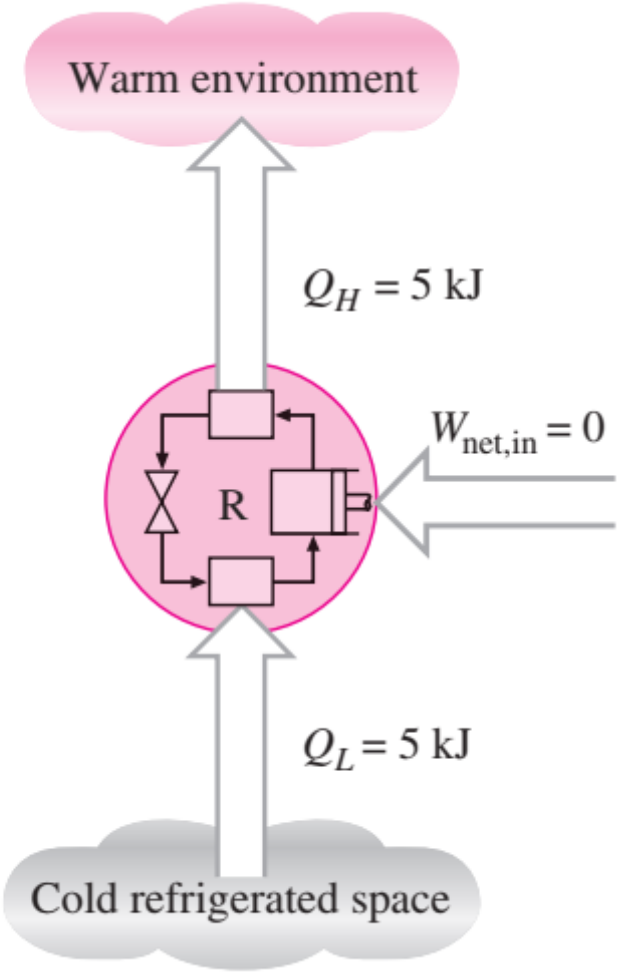
- ***Zeroth law of thermodynamics*** – If two thermodynamic systems are each in thermal equilibrium with a third, then they are in thermal equilibrium with each other.
- ***First law of thermodynamics*** – Energy can neither be created nor destroyed. It can only change forms. In any process, the total energy of the universe remains the same. For a thermodynamic cycle the net heat supplied to the system equals the net work done by the system.
- ***Second law of thermodynamics*** –
  - Kelvin–Planck Statement: It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work.

  - Clausius Statement: It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body.

Second Law of Thermodynamics



Kelvin–Planck Statement

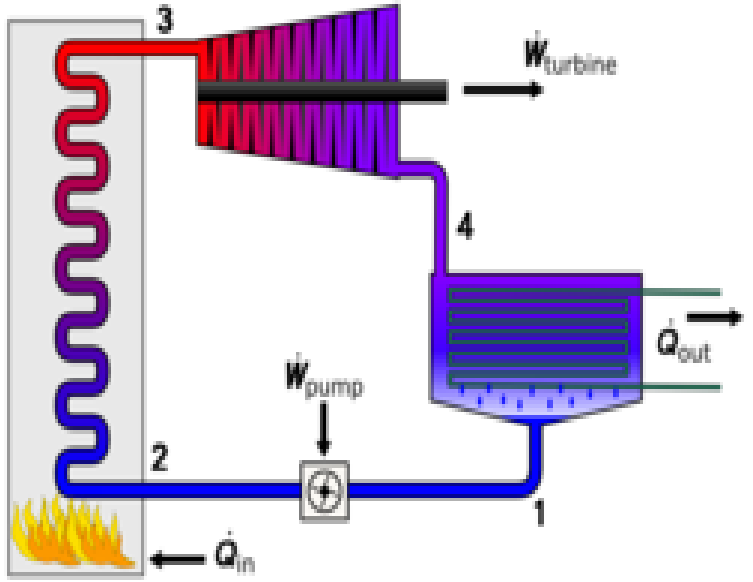


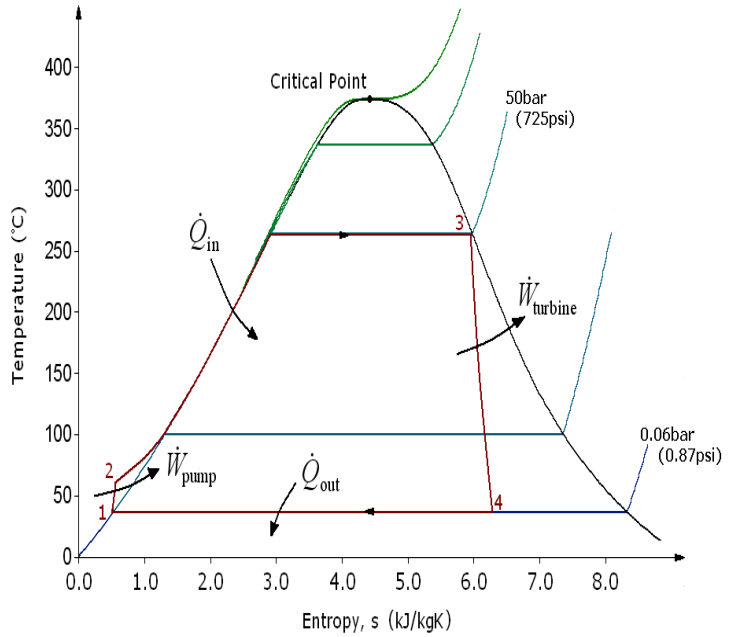
Clausius Statement

Rankine Cycle

the cycle undergoes a series of four processes: two isentropic (reversible adiabatic) processes alternated with two isobaric processes

1-2	Isentropic compression (compression in a centrifugal pump)	The liquid condensate is compressed adiabatically. The surroundings do work on the fluid, increasing its enthalpy (h=u+pv) and compressing it. S is constant. The work required for the compressor is given by W_{pump} = H2 – H1.
2-3	Isobaric heat addition (in heat exchanger – boiler)	constant-pressure. Heat transfer to the liquid from an external source. The net heat added is given by Qadd = H3 – H2.
3-4	Isentropic expansion (expansion in a turbine)	steam expands adiabatically. The steam does work on the surroundings (blades of the turbine) and loses an amount of enthalpy equal to the work that leaves the system. The work done by turbine is given by WT = H4 – H3. S is constant.
4-5	Isobaric heat rejection (in heat exchanger)	constant-pressure process. heat is rejected from the partially condensed gas. The net heat rejected is given by Qre = H4 – H1





Steam turbine

Power cycles normally with external combustion - or heat pump cycles:

Cycle	Compression, 1→2	Heat addition, 2→3	Expansion, 3→4	Heat rejection, 4→1	Notes
Bell Coleman	adiabatic	isobaric	adiabatic	isobaric	A reversed Brayton cycle
Carnot	isentropic	isothermal	isentropic	isothermal	Carnot heat engine
Ericsson	isothermal	isobaric	isothermal	isobaric	The second Ericsson cycle from 1853
Rankine	adiabatic	isobaric	adiabatic	isobaric	Steam engine
Hygroscopic	adiabatic	isobaric	adiabatic	isobaric	Hygroscopic cycle
Scuderi	adiabatic	variable pressure and volume	adiabatic	isochoric	
Stirling	isothermal	isochoric	isothermal	isochoric	Stirling engine
Manson	isothermal	isochoric	isothermal	isochoric then adiabatic	Manson-Guise Engine
Stoddard	adiabatic	isobaric	adiabatic	isobaric	

Power cycles normally with internal combustion:

Cycle	Compression, 1→2	Heat addition, 2→3	Expansion, 3→4	Heat rejection, 4→1	Notes
Brayton	adiabatic	isobaric	adiabatic	isobaric	Jet engine. The external combustion version of this cycle is known as first Ericsson cycle from 1833.
Diesel	adiabatic	isobaric	adiabatic	isochoric	Diesel engine
Lenoir	isobaric	isochoric	adiabatic		Pulse jets. Note, 1→2 accomplishes both the heat rejection and the compression.
Otto	adiabatic	isochoric	adiabatic	isochoric	Gasoline / petrol engines